CHAPTER 34

Mensuration II: Volume and Surface Area

INTRODUCTION

Solids

A *solid* is a figure bounded by one or more surfaces. It has three dimensions, namely, length, breath or width, and thickness or height. The plane surfaces that bind it are called its *faces*.

The *volume* of any solid figure is the amount of space enclosed within its bounding faces. It is measured in cubic units, e.g., m^3, cm^3, etc.

The area of the plane surfaces that bind the solid is called its *surface area*.

For any regular solid,

Number of faces + Number of vertices

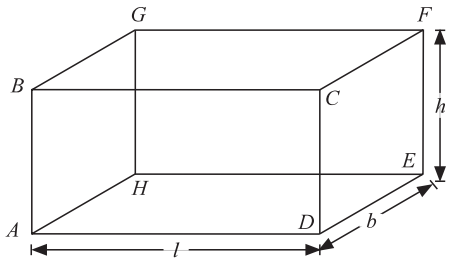
= Number of edges + 2.

We discuss here some important three-dimensional figures and the formulae associated with them.

Cubic

It is ta solid figure which has six rectangular faces. It is also called *rectangular parallelopiped.*

BASIC FORMULAE



If l, b and h denote the length, breadth and height of the cuboid, and d denotes the body diagonal (AF or BE or DG or CH), then

(i) Volume = $l \times b \times h = \sqrt{A_1 \times A_2 \times A_3}$,

where, A_1 = area of base or top,

A_2 = area of one side face, and

A_3 = area of other side face.

(ii) Total surface area = $2(lb + bh + lh)$

$= (l + b + h)^2 - d^2$

(iii) Diagonal of cuboid = $\sqrt{l^2 + b^2 + h^2}$

Notes

1. For painting the surface area of a box or to know how much tin sheet is required for making a box, we use formula (ii).
2. To find how much a box contains or how much space a box shall occupy, we use formula (i). To find the length of the longest pole to be placed in a room, we use formula (iii).
3. The rise or fall of liquid level in a container

$$= \frac{\text{Total volume of objects submerged or taken out}}{\text{Cross-sectional area of container}}.$$

Illustration 1: Find the volume and the total surface area of a cuboid whose dimensions are 25 m, 10 m and 2 m.

Solution: Here, l = 25 m, b = 10 m and h = 2 m.

Volume of the cuboid = $l \times b \times h$

$= 25 \times 10 \times 2$

$= 500\ m^3$

Total surface area of the cuboid

$= 2\ (lb + bh + lh)$

$= 2\ (25 \times 10 + 10 \times 2 + 25 \times 2)$

$= 2\ (250 + 20 + 50)$

$= 640\ \text{m}^2$.

Illustration 2: Find out the length of the longest bamboo that can be placed in a room which is 12 m long, 9 m broad and 8 m high.

Solution: Length of the bamboo

= length of the diagonal of the room

$= \sqrt{12^2 + 9^2 + 8^2}$

$= \sqrt{289} = 17$ m.

Illustration 3: The area of a side of a box is 120 cm^2. The area of the other side of the box is 27 cm^2. If the area of the upper surface of the box is 60 cm^2, then find out the volume of the box.

Solution: Volume to the box

$= \sqrt{\text{area of base} \times \text{area of one face} \times \text{area of the other face}}$

$= \sqrt{60 \times 120 \times 72}$

$= \sqrt{518400} = 720\ \text{cm}^3$.

Illustration 4: The sum of length, breadth and height of a cuboid is 12 cm. Find out the total surface area of the cuboid.

Solution: Total surface area

$= (\text{Sum of all three sides})^2 - (\text{Diagonal})^2$

$= 12^2 - 8^2 = 144 - 64 = 80\ \text{cm}^2$.

Cube

It is a special type of cuboid in which each face is a square.

For a cube, length, breadth and height are equal and is called, the edge of the cube,

If a be the edge of a cube, then

(i) Volume of the cube = $(\text{edge})^3 = a^3$

(ii) Total surface area of the cube = $6\ (\text{edge})^2 = 6a^2$

(iii) Diagonal of the cube = $\sqrt{3}a$ (edge) = $\sqrt{3}a$

(iv) Volume of the cube = $\left(\frac{\text{diagonal}}{\sqrt{3}}\right)^3 = \left(\frac{d}{\sqrt{3}}\right)^2$

$= \left(\sqrt{\frac{\text{Surface area}}{6}}\right)^3$

(v) Total surface area of the cube

$= 2\ (\text{diagonal})^2 = 2d^2$

(vi) For two cubes

(a) Ratio of volumes = $(\text{ratio of sides})^3$

(b) Ratio of surface areas = $(\text{Ratio of sides})^2$

(c) $(\text{Ratio of surface areas})^3 = (\text{Ratio of volumes})^2$.

Illustration 5: Find out the volume, surface area and the diagonal of a cube, each of whose sides measures 4 cm.

Solution: Volume of the cube = $a^3 = (4)^3 = 64\ \text{cm}^3$.

Surface area of the cube = $6a^2 = 6\ (4)^2 = 96\ \text{cm}^2$.

Diagonal of the cube = $\sqrt{3}a = 4\sqrt{3}$ cm.

Illustration 6: The surface area of a cube is 216 cm^2. Find out its volume.

Solution: Volume of the cube

$= \left(\sqrt{\frac{\text{Surface area}}{6}}\right)^3$

$= \left(\sqrt{\frac{216}{6}}\right)^3 = (6)^3 = 216\ \text{cm}^3$.

Illustration 7: The diagonal of a cube is $8\sqrt{3}$ cm. Find out its total surface area and volume.

Solution: We have,

Diagonal of cube = $\sqrt{3}$ (edge)

$\therefore$ Edge of cube = $\frac{\text{Diagonal of cube}}{\sqrt{3}}$

$= \frac{8\sqrt{3}}{\sqrt{3}} = 8$ cm.

Total surface area = $6\ (\text{edge})^2 = 6(8)^2$

$= 384\ \text{cm}^2$.

Volume of cube = $(\text{edge})^3 = (8)^3 = 512\ \text{cm}^3$.

Illustration 8: If the volumes of two cubical blocks are in the ratio of 8:1, then what will be the ratio of their edges?

Solution: We have,

Ratio of volumes = $(\text{Ratio of sides})^3$

Since, ratio of volumes = 8:1, i.e., $2^3:1^3$

$\therefore$ ratio of sides = 2:1.

Illustration 9: Volumes of the two cubes are in the ratio of 1:9. Find the ratio of their surface areas.

Solution: $(\text{Ratio of the surface areas})^3$

$= (\text{Ratio of volumes})^2$

$\therefore$ Ratio of surface areas = $\sqrt[3]{1:81} = 1:3\ (3)^{1/3}$.

Illustration 10: Sides of two cubes are in the ratio of 2:3. Find out the ratio of their surface areas.

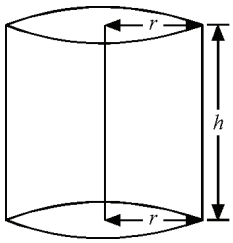
Solution: Ratio of surface areas

= (Ratio of sides)^2

= $(2:3)^2$ = 4:9.

Right Circular Cylinder

A *right circular cylinder* is a solid with circular ends of equal radius and the line joining their centres perpendicular to them. This is called, axis of the cylinder. The length of the axis is called, the height of the cylinder.



Notes

Take a rectangular sheet of paper and role it lengthwise or breadthwise in a round way, you will get a cylinder, i.e., a cylinder is generated by rotating a rectangle by fixing one of its sides.

If r is the radius of base and h is the height of the cylinder, then

(i) Volume of cylinder

= Area of the base × height

= $\pi r^2 \times h = \pi r^2 h$ cubic units

(ii) Area of the curved surface

= Circumference of the base × height

= $2\pi r \times h = 2\pi rh$ sq units

(iii) Area of the total surface

= Area of the curved surface

+ Area of the two circular ends

= $2\pi rh + 2\pi r^2$

= $2\pi r (h + r)$ sq units.

(iv) For two cylinders,

When radii are equal

(a) Ratio of volumes = Ratio of heights

(b) Ratio of volumes

= Ratio of curved surface areas

(c) Ratio of curved surface areas

= Ratio of heights

When heights are equal

(a) Ratio of volumes = (Ratio of radii)^2

(b) Ratio of volumes

= (Ratio of curved surface areas)^2

(c) Radii of curved surface areas

= Ratio of radii

When volumes are equal

(a) Ratio of radii = $\sqrt{\text{Inverse ratio of heights}}$

(b) Ratio of curved surface areas

= Inverse ratio of radii

(c) Ratio of curved surface areas

= $\sqrt{\text{Ratio of heights}}$

When curved surface areas are equal

(a) Ratio of radii = Inverse ratio of heights

(b) Ratio of volumes = Inverse ratio of heights

(c) Ratio of volumes = Ratio of radii

(v) For a cylinder

(a) Ratio of radii = (Ratio of curved surfaces) × (Inverse ratio of heights)

(b) Ratio of heights = (Ratio of curved surfaces) × (Inverse ratio of radii)

(c) Ratio of curved surfaces

= (Ratio of radii) × (Ratio of heights).

Illustration 11: The diameter of the base of a right circular cylinder is 28 cm and its height is 10 cm. Find out the volume and area of the curved surface of the cylinder.

Solution: Radius of the base = $\frac{28}{2}$ = 14 cm.

Volume of the cylinder = $\pi r^2 h$

= $\frac{22}{7} \times 14 \times 14 \times 10$

= 6160 cm^3.

Area of the curved surface = $2\pi rh$

= $2 \times \frac{22}{7} \times 14 \times 10$

= 880 cm^2.

Illustration 12: A cylinder of height 21 cm has base of radius 4 cm. Find out the total surface area of the cylinder.

Solution: Total surface area = $2\pi r (h + r)$

= $2 \times \frac{22}{7} \times 4 \times (21 + 4)$

= $\frac{4400}{7} = 628\ \frac{4}{7}$ cm^2.

Illustration 13: A rectangular piece of paper is 71 cm long and 10 cm wide. A cylinder is formed by rolling the paper along its breadth. Find out the volume of the cylinder. $\left[\text{Take } \pi = \frac{355}{113}\right]$

Solution: Circumference of the paper = Breadth of the paper

$\Rightarrow \quad 2\pi r = 10$

$\Rightarrow \quad r = \frac{10}{2\pi} = \frac{10 \times 113}{2 \times 355} = \frac{113}{71}$ cm.

As the length of the paper becomes the height of the cylinder,

$\therefore$ Volume of the cylinder

$= \pi r^2 l$

$= \frac{355}{113} \times \frac{113}{71} \times \frac{113}{71} \times 71 = 565 \text{ cm}^3.$

Illustration 14: Two circular cylinders of equal volume have their heights in the ratio of 9:16. Find out the ratio of their radii.

Solution: Ratio of radii $= \sqrt{\text{inverse ratio of heights}}$

$= \sqrt{16:9} = 4:3.$

Illustration 15: Two circular cylinders of equal volume have their heights in the ratio of 16:25. Find out the ratio of their curved surface areas.

Solution: Ratio of curved surface areas

$= \sqrt{\text{Ratio of heights}} = \sqrt{16:25} = 4:5.$

Illustration 16: Two circular cylinders of equal volume have their radii in the ratio of 4:9. Find out the ratio of their curved surface areas.

Solution: Ratio of curved surface areas

= inverse ratio of radii = 9:4.

Illustration 17: Two circular cylinders of equal heights have their radii in the ratio of 2:5. Find out the ratio of their volumes.

Solution: Ratio of volumes = (Ratio of radii)^2 = 4:25.

Illustration 18: Two circular cylinders of equal heights have their curved surface areas in the ratio of 3:5. Find out the ratio of their volumes.

Solution: Ratio of volumes

= (Ratio of curved surface areas)^2

= 9:25.

Illustration 19: Two circular cylinders of equal curved surface areas have their heights in the ratio of 4:7. Find out the ratio of their volumes.

Solution: Ratio of volumes = Inverse ratio of heights

$= \frac{1}{4} : \frac{1}{7} = 7:4.$

Illustration 20: Two circular cylinders of equal curved surface areas have their heights in the ratio of 4:5. Find out the ratio of their volumes.

Solution: Ratio of volumes = Inverse ratio of heights

$= \frac{1}{4} : \frac{1}{5} = 5:4.$

(vi) If the ratio of heights and the ratio of radii of two right circular cylinders are given, then

Ratio of curved surface areas = (ratio of radii) (ratio of heights).

Illustration 21: If the heights and the radii of two right circular cylinders are in the ratio 2:3 and 4:5, respectively. Find out the ratio of their curved surface areas.

Solution: Ratio of curved surface areas = (ratio of radii) (ratio of heights)

= (4:5) (2:3) = 8:15.

(vii) If the ratio of heights and the ratio of curved surface areas of two right circular cylinders are given, then

Ratio of radii = (ratio of curves surface areas) (inverse ratio of heights).

Illustration 22: The heights and curved surface areas of two right circular cylinders are in the ratio 3:4 and 5:8, respectively. Find out the ratio of their radii.

Solution: Ratio of radii = (ratio of curved surface areas) (inverse ratio of heights)

$= (5:8) \left(\frac{1}{3} : \frac{1}{4}\right) = (5:8)\ (4:3) = 5:6.$

(viii) If the ratio of radii and the ratio of curved surface areas of two right circular cylinders are given, then

Ratio of heights = (ratio of curved surface areas) (inverse ratio of radii)

Illustration 23: The radii of two right circular cylinders are in the ratio of 3:4 and their curved surface areas are in the ratio of 5:6. Find out the ratio of their heights.

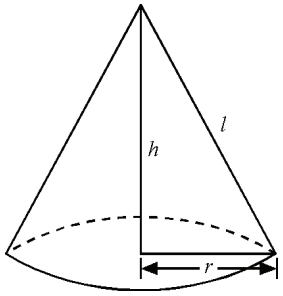
Solution: Ratio of heights = (ratio of curved surface areas) (inverse ratio of radii)

$= (5:6) \left(\frac{1}{3} : \frac{1}{4}\right)$

$= (5:6)\ (4:3) = 10:9.$

Right Circular Cone

A *right circular cone* is a solid obtained by rotating a right-angled triangle around its height.



If r = radius of base; h = height,

l = slant height = $\sqrt{h^2 + r^2}$, then

(i) Volume of cone

$= \frac{1}{3} \times$ area of the base $\times$ height

$= \frac{1}{3} \times \pi r^2 h$ cubic units

(ii) Area of curved surface = πrl

$= \pi r\sqrt{h^2 + r^2}$ sq. units

(iii) Total surface area of cone

= Area of the base + area of the curved surface

$= \pi r^2 + \pi rl = \pi r (r + l)$ sq units.

(iv) For two cones

(a) When volumes are equal

Ratio of radii = $\sqrt{\text{inverse ratio of heights}}$

(b) When radii are equal

Ratio of volumes = Ratio of heights

(c) When heights are equal

Ratio of volumes = (ratio of radii)^2

(d) When curved surface areas are equal

Ratio of radii = inverse ratio of slant heights.

Illustration 24: Find out the slant heights of a cone whose volume is 1232 cm^3 and radius of the base is 7 cm.

Solution: Volume of the cone = $\frac{1}{3}\pi r^2 h = 1232$

$\Rightarrow \quad h = \frac{1232 \times 3}{\pi r^2} = \frac{1232 \times 3 \times 7}{22 \times 7 \times 7} = 24$ cm.

Slant height l is given by the relation

$l = \sqrt{h^2 + r^2}$

$= \sqrt{(24)^2 + (7)^2} = \sqrt{576 + 49}$

$= \sqrt{625} = 25$ cm.

$\therefore$ Slant height of the cone is 25 cm.

Illustration 25: A tent is of diameter 12 m at the base and its height is 8 m.

(i) Find the slant height; and

(ii) The canvas required in m^2.

How many persons can the tent accommodate, at the most, if each person requires 18 m^3 of air?

Solution: Diameter of the base of a conical tent = 12 m.

$\therefore$ Radius (r) = $\frac{12}{2}$ = 6 m and its height (h) = 8 m.

(i) Slant height (l) = $\sqrt{r^2 + h^2} = \sqrt{6^2 + 8^2}$

$= \sqrt{64 + 36} = \sqrt{100}$ = 10 m.

(ii) Area of canvas required

$= \pi \times r \times l$

$= \frac{22}{7} \times 6 \times 10 = 188.57$ m

(iii) Volume of conical portion

$= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times \frac{22}{7} \times 6 \times 6 \times 8 = 301.71$ m^3.

Space required for each person = 18 m^3.

$\therefore$ Number of persons that can be accommodated

$= \frac{301.71}{18} = 16.$

Illustration 26: The height of a cone is 21 cm and radius of its base is 28 cm. Find out its total surface area.

Solution: We have, r = 28 cm and h = 21 cm.

Slant height (l) = $\sqrt{r^2 + h^2} = \sqrt{(28)^2 + (21)^2}$

$= \sqrt{1225} = 35$ cm.

Total surface area = $\pi r (l + r)$

$= \frac{22}{7} x \ 28 \times (35 + 28)$

$= \frac{22}{7} \times 28 \times 63$

$= 5544$ cm^2.

Illustration 27: Two right circular cones of equal curved surface areas have their slant heights in the ratio of 3:5. Find out the ratio of their radii.

Solution: Ratio of radii = inverse ratio of slant heights

$= \frac{1}{3} : \frac{1}{5} = 5:3.$

Illustration 28: Two right circular cones of equal volumes have their heights in the ratio of 4:9. Find out the ratio of their radii.

Solution: Ratio of radii $= \sqrt{\text{inverse ratio of heights}}$

$$= \sqrt{\frac{1}{4}:\frac{1}{9}} = \sqrt{9:4} = 3:2.$$

Illustration 29: Two right circular cones of equal heights have their radii in the ratio of 1:3. Find out the ratio of their volumes.

Solution: Ratio of volumes = (Ratio of radii)^2

$$= (1:3)^2 = 1:9.$$

(v) If the ratio of volumes and the ratio of heights of two right circular cones (or cylinders) are given, then

Ratio of radii

$= \sqrt{(\text{ratio of volumes})(\text{inverse ratio of heights})}$

$= \sqrt{(3:2)(8:3)} : \sqrt{4:1} = 2:1.$

Illustration 30: The volumes of two cones are in the ratio 3:2 and their heights in the ratio 3:8. Find out the ratio of their radii.

Solution: Ratio of radii

$= \sqrt{(\text{ratio of volumes})(\text{inverse ratio of heights})}$

$= \sqrt{(3:2)(8:3)} = \sqrt{4:1} = 2:1.$

(vi) If the ratio of heights and the ratio of diameters (or radii) of two right circular cones (or cylinders) are given, then

Ratio of volumes = (ratio of radii)^2 × (ratio of heights).

Illustration 31: The heights of two cones are in the ratio of 5:3 and their radii is in the ratio 2:3. Find out the ratio of their volumes.

Solution: Ratio of volumes

$= (\text{ratio of radii})^2 \times (\text{ratio of heights})$

$= (2:3)^2 \times (5:3)$

$= \frac{4}{9} \times \frac{5}{3} = 20:27.$

(vii) If the ratio of radii (or diameter) and the ratio of volumes of two right circular cones are given, then ratio of heights

= (inverse ratio of radii)^2 (ratio of volumes).

Illustration 32: The volumes of two cones are in the ratio of 1:4 and their diameters are in the ratio of 4:5. Find out the ratio of their heights.

Solution: Ratio of heights

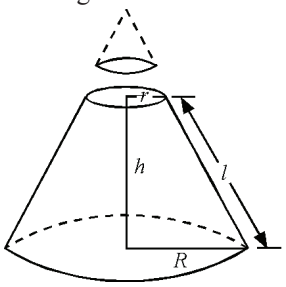
$= (\text{inverse ratio of diameters})^2 \times (\text{ratio of volumes})$

$= \left(\frac{1}{4}:\frac{1}{5}\right)^2 (1:4) = (5:4)^2\ (1:4)$

$= \frac{25}{16} \times \frac{1}{4} = 25:64.$

Frustum of a Right Circular Cone

A cone with some of its top portion cut off is called, the *frustum* of the original cone.



If R = Radius of the base of frustum
r = Radius of the top of the frustum
h = Height of the frustum
l = Slant height of the frustum, then

(a) Slant height = $\sqrt{h^2 + (R-r)^2}$ units

(b) Area of the curved surface = $\pi\ (R + r)\ l$ sq. units

(c) Total surface area of the frustum
$= \pi\ [(R^2 + r^2) + l\ (R + r)]$ sq. units

(d) Volume of the frustum = $\frac{\pi h}{3}(R^2 + r^2 + Rr)$ cu units.

Illustration 33: A reservoir is in the shape of a frustum of a right circular cone. It is 8 m across at the top and 4 m across the bottom. It is 6 m deep. Find out the area of its curved surface, total surface area and also its volume.

Solution: Here, $R = 4$, $r = 2$ and $h = 6$.

$\therefore$ Slant height $(l) = \sqrt{h^2 + (R-r)^2}$

$= \sqrt{(6)^2 + (4-2)^2} = \sqrt{40}$.

$\therefore$ Area of the curved surface

$= \pi\ (R + r)\ l$

$= \frac{22}{7}(4 + 2)\sqrt{40}$

$= 18.8 \times 6.3 = 118.4$ m

Total surface area $= \pi\ [(R^2 + r^2) + l\ (R + r)]$

$= \frac{22}{7}[(4^2 + 2^2) + \sqrt{40}\ (4 + 2)]$

$= \frac{22}{7}(20 + 6\sqrt{40}) = 181.6\ \text{m}^2$

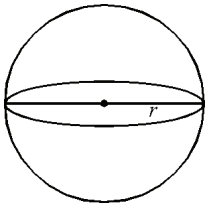
Volume of the frustum $= \frac{\pi h}{3}(R^2 + r^2 + Rr)$

$= \frac{22}{7} \times \frac{6}{3}(4^2 + 2^2 + 4 \times 2)$

$= \frac{44}{7}(16 + 4 + 8) = 176\ \text{m}^3.$

Sphere

A *sphere* is the solid figure formed by revolving a semi-circle on its diameter.

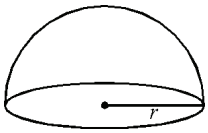


The mid-point of the diameter is called, centre of the sphere, and the radius of the semi-circle is called, the radius of the sphere.

If r = radius of the spheres, then

(i) Volume of sphere = $\frac{4}{3}\pi r^3$ cubic units

(ii) Surface area = $4\pi r^2$ sq units.



(iii) Volume of hemisphere = $\frac{2}{3}\pi r^2$ cubic units

(iv) Area of curved surface = $2\pi r^2$ sq units of hemisphere

(v) Total surface area of hemisphere = $3\pi r^2$ sq units.

Illustration 34: Diameter of a sphere is 28 cm. Find out its surface area and volume.

Solution: Radius of the sphere (r) = $\frac{28}{2}$ = 14 cm.

$$\text{Surface area} = 4\pi r^2 = 4 \times \frac{22}{7} \times 14 \times 14$$
$$= 2464 \text{ cm}^2.$$

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3$$
$$= \frac{4}{3} \times \frac{22}{7} \times 14 \times 14 \times 14$$
$$= 11498.6 \text{ cm}^3.$$

Illustration 35: Find out the volume, curved surface area and total surface area of a hemisphere of radius 21 cm.

Solution: Volume of the hemisphere

$$= \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 21 \times 21 \times 21 = 19494 \text{ cm}^3.$$

Curved surface area = $2\pi r^2$

$$= 2 \times \frac{22}{7} \times 21 \times 21$$
$$= 2772 \text{ cm}^2.$$

Total surface area = $3\pi r^2$

$$= 3 \times \frac{22}{7} \times 21 \times 21$$
$$= 4158 \text{ cm}^2.$$

(vi) For two spheres

(a) (Ratio of radii)^2 = Ratio of surface areas

(b) (Ratio of radii)^3 = Ratio of volumes

(c) (Ratio of surface areas)^3 = (Ratio of volumes)^2.

Illustration 36: The radii of two spheres are in the ratio of 2:3. What is the ratio of their surface areas?

Solution: Ratio of surface areas = (ratio of radii)^2
$= (2:3)^2 = 4:9.$

Illustration 37: The surface areas of two spheres are in the ratio 1:2. Find out the ratio of their volumes.

Solution: We have,

(Ratio of surface areas)^3 = (Ratio of volumes)^2

$\Rightarrow$ $(1:2)^3$ = (Ratio of volumes)^2

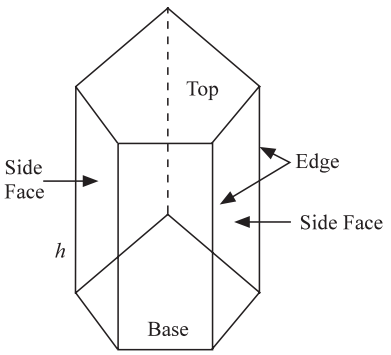
$\therefore$ Ratio of volumes = $\sqrt{1:8} = 1:2\sqrt{2}$.

Illustration 38: The radii of two spheres are in the ratio of 2:45. Find out the ratio of their volumes.

Solution: Ratio of volumes = (Ratio of radii)^3
$= (2:5)^3 = 8:125.$

Prism

A solid having top and bottom faces identical and side faces rectangular is a prism.



In a prism with a base of n sides;

Number of vertices = $2n$

and Number of faces = $n + 2$.

Volume of the prism = area of base × height

Lateral surface area = perimeter of base × height

Total surface area = 2 × Base area + Lateral Surface area.

Illustration 39: Find out the volume and the total surface area of a triangular prism whose height is 30 m and the sides of whose base are 21 m, 20 m and 13 m, respectively.

Solution: Perimeter of base = 21 + 20 + 13 = 54 m.
height = 30 m.

$$\text{Area of base} = \sqrt{s(s-a)(s-b)(s-c)}$$
$$= \sqrt{27(27-21)(27-20)(27-13)}$$
$$= \sqrt{27\times6\times7\times14} = 126 \text{ m}^2.$$

$\therefore$ Volume of the prism = area of base × height
= 126 × 54 = 6804 m^3.

Also, surface area of the prism
= 2 × Base area + lateral surface area
= 2 Base area + perimeter of base × height
= 2 × 126 + 54 × 30 = 1872 m^2.

SOLIDS INSCRIBED/CIRCUMSCRIBING OTHER SOLIDS

1. If a largest possible sphere is circumscribed by a cube of edge 'a' cm, then the radius of the sphere $= \frac{a}{2}$.

Illustration 40: Find out the volume of largest possible sphere circumscribed by a cube of edge 8 cm.

Solution: Radius of the sphere $= \frac{a}{2} = \frac{8}{2} = 4$ cm.

$\therefore$ Volume of the sphere $= \frac{4}{3}\pi r^3$

$$= \frac{4}{3}\times\frac{22}{7}\times4\times4\times4$$
$$= 268.1 \text{ cm}^3.$$

2. If a largest possible cube is inscribed in a sphere of radius 'a' cm, then the edge of the cube $= \frac{2a}{\sqrt{3}}$.

Illustration 41: Find out the surface area of largest possible cube inscribed in a sphere of radius 4 cm.

Solution: Edge of the cube $\frac{2a}{\sqrt{3}} = \frac{2\times4}{\sqrt{3}} = \frac{8}{\sqrt{3}}$.

$\therefore$ Surface area of the cube = 6 (edge)^2

$$= 6 \times \frac{64}{3}$$
$$= 128 \text{ cm}^2.$$

3. If a largest possible sphere is inscribed in a cylinder of radius 'a' cm and height 'h' cm, then

$$\text{radius of the sphere} = \begin{cases} a \text{ for } h > a \\ \frac{h}{2} \text{ for } a > h \end{cases}$$

Illustration 42: Find out the surface area of largest possible sphere inscribed in a cylinder of radius 14 cm and height 17 cm.

Solution: Radius of the sphere = 14 cm ($\because$ $h > a$)

$\therefore$ Surface area of sphere $= 4\pi r^2$

$$= 4 \times \frac{22}{7}\times14\times14$$
$$= 2464 \text{ cm}^2.$$

4. If a largest possible sphere is inscribed in a cone of radius 'a' cm and slant height equal to the diameter of the base, then radius of the sphere $= \frac{a}{\sqrt{3}}$.

Illustration 43: Find out the surface area of largest possible sphere inscribed in a cone of radius 21 cm and slant height equal to the diameter of the base.

Solution: Radius of the sphere $= \frac{a}{\sqrt{3}} = \frac{21}{\sqrt{3}}$ cm.

$\therefore$ Surface area of the sphere $= 4\pi r^2$

$$= 4 \times \frac{22}{7}\times\frac{21}{\sqrt{3}}\times\frac{21}{\sqrt{3}}$$
$$= 1848 \text{ cm}^2.$$

5. If a largest possible cone is inscribed in a cylinder of radius 'a' cm and height 'h' cm, then radius of the cone = a and height = h.

Illustration 44: Find out the volume of largest possible cone inscribed in a cylinder of radius 6 cm and height 14 cm.

Solution: Radius of the cone (r) = 6 cm.
and height of the cone (h) = 14 cm.

$\therefore$ Volume of the cone $= \frac{1}{3}\pi r^2 h$

$$= \frac{1}{3}\times\frac{22}{7}\times6\times6\times14$$
$$= 528 \text{ cm}^2.$$

6. If a largest possible cube is inscribed in a hemisphere of radius 'a' cm, then the edge of the cube = $a\sqrt{\frac{2}{3}}$.

Illustration 45: Find out the length of the diagonal of largest possible cube inscribed in a hemisphere of radius $4\sqrt{2}$ cm.

Solution: Edge of the cube $= a\sqrt{\frac{2}{3}} = 4\sqrt{2} \times \sqrt{\frac{2}{3}} = \frac{8}{\sqrt{3}}$ cm.

$\therefore$ Diagonal of the cube = $\sqrt{3}$ (edge)

$= \sqrt{3} \times \frac{8}{\sqrt{3}} = 8$ cm.

SHORT-CUT METHODS

01 If all three measuring dimensions of a sphere, cuboid, cube, cylinder or cone are increased or decreased by x%, y% and z% respectively, then the volume of the figure will increase or decrease by

$$\left(x + y + z + \frac{xy + yz + zx}{100} + \frac{xyz}{100^2}\right)\%$$

For cuboid, the three measuring dimensions are length, breadth and height.

For cube, all three measuring dimensions are equal, i.e., $x = y = z$.

For sphere also, (or diameter) all three measuring dimensions are equal and is given by radius, i.e., $x = y = z = r$.

For cylinder or a cone two measuring dimensions are equal to radius and third measuring dimension is height

i.e., $x = y = r$ and $z = h$.

Illustration 46: The length, breadth and height of a cuboid are increased by 5%, 10% and 20%, respectively. Find out the percentage increase in its volume.

Solution: Here, $x = 5$, $y = 10$ and $z = 20$.

$\therefore$ Percentage increase in volume

$$= \left[x + y + z + \frac{xy + xz + yz}{100} + \frac{xyz}{(100)^2}\right]\%$$

$$= \left[5 + 10 + 20 + \frac{(5\times10) + (5\times20) + (10\times20)}{100} + \frac{5\times10\times20}{(100)^2}\right]\%$$

$$= \left(35 + \frac{350}{100} + \frac{1000}{(100)^2}\right)$$

$= (35 + 3.5 + 0.1)\%$

$= 38.6\%$

Illustration 47: The sides of a cube are decreased by 10% each. Find out the percentage change in its volume.

Solution: Here, $x = y = z$.

$\therefore$ Percentage change in volume

$$= \left[3x + \frac{3x^2}{100} + \frac{x^3}{(100)^2}\right]\%$$

$$= \left[3(-10) + \frac{3(-10)^2}{100} + \frac{(-10)^3}{(100)^2}\right]\%$$

$= (-30 + 3 - 0.1)\% = -27.1\%$

−ve sign indicates decrease in volume, that is, there is a decrease in volume by 27.1%

Illustration 48: The diameter of a sphere is increased by 20%. What is the percentage increase in its volume?

Solution: Percentage increase in volume

$$= \left[3x + \frac{3x^2}{100} + \frac{x^3}{(100)^2}\right]\% \quad [\text{Here, } x = y = z]$$

$$= \left[3\times20 + \frac{3(20)^2}{100} + \frac{(20)^3}{(100)^2}\right]\%$$

$= (60 + 12 + 0.8)\% = 72.8\%$

Illustration 49: The radius of a right circular cylinder is decreased by 5%, but its height is increased by 10%. What is the percentage change in its volume?

Solution: Here, $x = y = -15$ and $z = 10$.

$\therefore$ Percentage change in volume

$$= \left[-5 - 5 + 10 + \frac{(-5)(-5) + (-5)(10) + (-5)(10)}{100} + \frac{(-5)(-5)(10)}{(100)^2}\right]\%$$

$= (0 - 0.75 + 0.025)\% = -0.725\%$

Therefore, volume decrease by 0.725%

Illustration 50: Each of the radius and the height of a cone is increased by 25%. Find out the percentage increase in volume.

Solution: Here, $x = y = 25$ and $z = 25$.

$\therefore$ Percentage increase in volume

$$= \left[25+25+25+\frac{25\times25+25\times25+25\times25}{100}+\frac{25\times25\times25}{(100)^2}\right]\%$$

$= (75 + 18.75 + 1.56)\% = 95.3\%$

02 If the two measuring dimensions which are included in the surface area of a sphere, cuboid, cube, cylinder or cone are increased or decreased by x% and y%, then the surface area of the figure will increase or decrease by

$$\left(x+y+\frac{xy}{100}\right)\%$$

Note that in case of percentage increase, values of x, y and z are positive, and in case of percentage decrease, values of x, y and z are negative.

Illustration 51: Each edge of a cube is increased by 20%. What is the percentage increase in its surface area?

Solution: Here, $x = y = 20$.

$\therefore$ Percentage increase in surface area

$$= \left(x+y+\frac{xy}{100}\right)\%$$

$$= \left(20+20+\frac{20\times20}{100}\right)\%$$

$= (40 + 4)\% = 44\%$

Illustration 52: The radius of a hemisphere is decreased by 10%. Find out the percentage change in its surface area.

Solution: Here, $x = y = -10$.

$\therefore$ Percentage change in surface area

$$= \left(x+y+\frac{xy}{100}\right)\%$$

$$= \left(-10-10+\frac{(-10)(-10)}{100}\right)\%$$

$= (-20 + 1)\% = -19\%$

Therefore, surface area of hemisphere decreases by 19%

Illustration 53: The radius of a right circular cone is increased by 25% and slant height is decreased by 30%. Find out the percentage change in curved surface area of the cone.

Solution: Here, $x = 25$ and $y = -30$.

$\therefore$ Percentage change in curved surface area

$$= \left(x+y+\frac{xy}{100}\right)\%$$

$$= \left[25-30+\frac{(25)(-30)}{100}\right]\%$$

$= (-5 - 7.5)\% = -12.5\%$

Therefore, curved surface area decreases by 12.5%

Illustration 54: The radius and height of a cylinder are increased by 10% and 20%, respectively. Find out the percentage increase in its surface area.

Solution: Here, $x = 10$ and $y = 20$.

$\therefore$ Percentage increase in surface area

$$= \left(x+y+\frac{xy}{100}\right)\%$$

$$= \left(10+20+\frac{10\times20}{100}\right)\% = 32\%$$

03 If a sphere of radius R is melted to form smaller spheres each of radius r, then

The number of smaller spheres

$$= \frac{\text{Volume of the bigger sphere}}{\text{Volume of the smaller sphere}}$$

$$= \left(\frac{R}{r}\right)^3.$$

Illustration 55: Find out the number of lead balls of radius 1 cm each that can be made from a sphere of radius 4 cm.

Solution: Number of lead balls $= \left(\frac{R}{r}\right)^3 = \left(\frac{4}{1}\right)^3 = 64.$

04 If by melting n spheres, each of radius r, a big sphere is made, then

Radius of the big sphere $= r\cdot\sqrt[3]{n}$.

Illustration 56: If by melting 8 spheres, each of radius 5 cm, a big sphere is made, what will be the radius of the big sphere?

Solution: Radius of the big sphere $= r\cdot\sqrt[3]{n}$

$= 5\cdot\sqrt[3]{8} = 5\cdot2 = 10$ cm.

05 If a cylinder is melted to form smaller spheres each of radius r, then

The number of small spheres $= \frac{\text{Volume of cylinder}}{\text{Volume of 1 sphere}}$.

Illustration 57: How many bullets can be made out of a loaded cylinder 24 cm high and 5 cm diameter, each bullet being 2 cm in diameter?

Solution: Number of bullets = $\frac{\text{Volume of cylinder}}{\text{Volume of 1 sphere}}$.

$$= \frac{\pi \times \frac{5}{2} \times \frac{5}{2} \times 24}{\frac{4}{3} \times \pi \times 1 \times 1 \times 1} = 450.$$

06 If a sphere of radius r is melted and a cone of height h is made, then

Radius of the cone = $2 \times \sqrt{\frac{r^3}{h}}$.

or,

If a cone of height h is melted and a sphere of radius r is made, then

Radius of the cone = $2 \times \sqrt{\frac{r^3}{h}}$.

Illustration 58: A solid cone of copper of height 3 cm is melted and a solid sphere of radius 3 cm is made. Then what is the diameter of the base of the cone?

Solution: Radius of the base of the cone

$$= 2 \times \sqrt{\frac{r^3}{h}} = 2 \times \sqrt{\frac{3^3}{3}} = 6$$

$\therefore$ Diameter of the base of the cone = 2×6

= 12 cm.

Illustration 59: If a solid cone of copper of height 2 cm is melted and a solid sphere of radius 2 cm is made, what is the diameter of the base of the cone?

Solution: Radius of the base of the cone

$$= 2 \times \sqrt{\frac{r^3}{h}} . = 2 \times \sqrt{\frac{(2)^3}{2}}$$

$= 2 \times 2 = 4$ cm.

$\therefore$ Diameter of the base of the cone = $2 \times 4 = 8$ cm.

EXERCISE-I

1. A tank, 16 m long and 23 m wide contains water. How many cubic m of water must be rushed into it to make the surface rise by $16\frac{2}{3}$ cm?

(a) 48 m^3 (b) 40 m^3
(c) 32 m^3 (d) 42 m^3

2. The outer dimensions of a closed box are 12 cm by 10 cm by 8 cm. If the box is made of wood 1 cm thick, find out the capacity of the box.

(a) 360 cm^3 (b) 480 cm^3
(c) 240 cm^3 (d) 560 cm^3

3. A cistren of dimensions 2.4 m × 2.0 m × 1.5 m takes 2 hours 30 minutes to get filled with water. The rate at which water flows into the cistern is:

(a) 0.48000 cu.m/h (b) 800 cu.m/min
(c) 800 cu.m/sec (d) None of these

4. The area of three adjacent faces of a rectangular box are p, q and r square cm. The volume of the box is given by:

(a) $(p + q + r)$ cm^3 (b) $\sqrt{pqr}$ cm^3
(c) $(pqr)^{1/3} cm^3$ (d) pqr cm^3

5. A reservoir, 30 m long, and 15 m broad, is filled with water. How many gallons of water must be taken out to lower the level of water by 4 m?

(a) 342000 gallons (b) 364200 gallons
(c) 324000 gallons (d) 386400 gallons

6. How many bricks, each measuring 250 cm by 12.5 cm by 7.5 cm, will be required to build a 5 m long, 3 m high and 20 cm thick wall?

(a) 1480 (b) 1280
(c) 1680 (d) 1480

7. Find out the cost of the log of wood measuring $15\frac{1}{2}$ m by $2\frac{3}{4}$ m by $1\frac{1}{3}$ m at ₹45 per cm^3.

(a) ₹4257.50 (b) ₹4005.00
(c) ₹4207.50 (d) ₹4357.50

8. How many bricks are required to build a 15 m long 3 m high and 50 cm thick wall, if each brick measures 25 cm by 12 cm by 6 cm.

(a) 16500 (b) 14500
(c) 12500 (d) 10500

9. Find the diagonal of a cuboid whose dimensions are 12 m by 10 m by 8 m.

(a) 18 m (b) 17.5 m
(c) 17 m (d) 16.5 m

10. The outer dimensions of a closed wooden box of 1 cm thick are 12 cm by 10 cm by 8 cm. Find out the cost of the wood required to make the box if 1 cm^3 of wood costs ₹3.00.

(a) ₹1440 (b) ₹1640
(c) ₹1840 (d) ₹2040

11. 3 equal cubes are placed adjacently in a row. Find out the ratio of the total surface area of the new cuboid to that of the sum of the surface areas of the three cubes:

(a) 3:5 (b) 4:5
(c) 6:7 (d) 7:9

12. The diagonal of a cubical box is $\sqrt{300}$ cm. Find out the surface area:

(a) $600\sqrt{3}$ cm^2 (b) 600 cm^2
(c) 1200 cm^2 (d) $900\sqrt{3}$ cm^2

13. An iron cube of 10 cm sides is hammered into a rectangular sheet of thickness 0.5 cm. If the sides of the sheet be in the ratio 1:5, the sides (in cm) are:

(a) 110 cm, 50 cm (b) 20 cm, 100 cm
(c) 40 cm, 200 cm (d) None of these

14. The length of a room is 12 m, width 8 m, and height 6 m. How many boxes will it hold if each is allowed 1.5 cubic metre of space?

(a) 864 (b) 506
(c) 384 (d) 436

15. A 3.3 m high room is half as long again as it is wide and its volume is $123\frac{3}{4}$ m^3. Find out its length and breadth.

(a) 7.5 m, 6 m (b) 8 m, 5 m
(c) 7.5 m, 5 m (d) 8.5 m, 5 m

16. A tank 3 m long, 2 m wide and 1.5 m deep is dug in a field 22 m long and 14 m wide. If the earth dug out is evenly spread out over the field, the level of the field will rise by nearly:

(a) 0.299 cm (b) 0.29 mm
(c) 2.98 cm (d) 4.15 cm

17. A school room is to be built to accommodate 70 children, so as to allow 2.2 m^2 of floor and 11 m^3 of space for each child. If the room be 14 m long, what must be its breadth and height?

(a) 12 m, 5,5 m (b) 11 m, 5 m
(c) 13 m, 6 m (d) 11 m, 4 m

18. If 210 m^3 of sand be thrown into a tank 12 m long and 5 m wide, find how much the water will rise?

(a) 3.5 m (b) 4 m
(c) 7 m (d) Data inadequate

19. If the length, breadth and height of a rectangular parallelopiped are in the ratio 6:5:4 and if total surface area is 33,300 m^2, then the length, breadth and height of parallelopiped (in cm) respectively are:

(a) 90, 85, 600 (b) 90, 75, 70
(c) 85, 75, 60 (d) 90, 75, 60

20. A m^3 of metal weighing 90 Kg is rolled into a square bar 9 metre long. An exact cube is cut off from the bar. How much does it weigh?

(a) $5\frac{2}{3}$ Kg (b) $6\frac{1}{3}$ Kg
(c) $3\frac{1}{3}$ Kg (d) $4\frac{2}{3}$ Kg

21. How many cubes, each of surface 24 cm^2, can be made out of a cube of edge measure 1 metre?

(a) 165000 (b) 125000
(c) 180000 (d) 155000

22. 3 solid cubes whose edges are 6, 8 and 10 cm respectively, are melted and formed into a single cube. If there be no loss of metal in the process, find out the edge of the new cube.

(a) 16 cm (b) 10 cm
(c) 14 cm (d) 12 cm

23. If a cube with its edge 6 cm is melted and smaller cubes with edge 2 cm each are formed, then how many cubesare formed?

(a) 39 (b) 24
(c) 27 (d) 21

24. How many small cubical blocks of side 5 cm can be cut from a cubical block whose each edge measures 20 cm?

(a) 56 (b) 48
(c) 64 (d) 52

25. Surface area of a cube is 600 cm^2. Find out the length of its diagonal.

(a) $15\sqrt{3}$ (b) $12\sqrt{3}$
(c) $10\sqrt{3}$ cm (d) None of these

26. A rectangular tank is 30 m long and 20 m broad. Water is being flown into it through a square pipe of side 5 cm. What is the speed of water if the level of water in the tank rises by 1 m in 8 hours?

(a) 30 Km/h (b) 36 Km/h
(c) 40 Km/h (d) None of these

27. Calculate the number of bricks, each measuring 25 cm × 15 cm × 8 cm, required to construct a wall with its dimension 19 m × 4 m × 5 m, when 10% of its volume is occupied by mortar.

(a) 4000 (b) 8000
(c) 7000 (d) 6000

28. 3 cubes of metal whose edges are in the ratio 3:4:5 are melted into a single cube, the length of whose

diagonal is $48\sqrt{3}$ m. Calculate the edges of the three cubes.

(a) 24 m, 32 m, 40 m (b) 40 m, 32 m, 24 m
(c) 30 m, 22 m, 18 m (d) 48 m, 36 m, 24 m

29. A cube of lead with edges measuring 6 cm each is melted and recasted into 27 equal cubes. The length of the edge of the new cube is:

(a) 3 cm (b) 4 cm
(c) 2 cm (d) 1.5 cm

30. The volume of a cube is 729 cm^3. The total surface area of the cube is:

(a) 216 cm^2 (b) 384 cm^2
(c) 486 cm^2 (d) 512 cm^2

31. 2 cubes have volumes in the ratio 1:27. The ratio of the area of the face of one to that of the other is:

(a) 1:2 (b) 1:3
(c) 1:6 (d) 1:9

32. 2 cubes, each of side 12 cm, are joined end-to-end. The surface area of the resulting cuboid is:

(a) 1240 cm^2 (b) 1440 cm^2
(c) 2250 cm^2 (d) 4252 cm^2

33. The perimeter of one face of a cube is 20 cm. Its volume is:

(a) 1009 cm^3 (b) 525 cm^3
(c) 320 cm^3 (d) 125 cm^3

34. A cube of edge 3 cm of iron weighs 12 gm. What is the weight of a similar cube of iron whose edge is 12 cm?

(a) 768 gm (b) 678 gm
(c) 964 gm (d) 864 gm

35. The weight of a solid cube of iron of 1 cm edge is 17 gm. What should be the weight with a similar cube of edge 3 cm?

(a) 449 gm (b) 459 gm
(c) 469 gm (d) 4390 gm

36. A cubic metre of silver weighing 900 Kg is rolled into a 16 m long square bar. Find out the weight of an exact cube cut off from it.

(a) 14 Kg $62\frac{1}{2}$ gm (b) 30 Kg
(c) 10 Kg (d) 7 Kg 50 gm

37. A 4 cm cube is cut into 1 cm cubes. What is the ratio of the surface area of small cubes to that of the large cube?

(a) 1:16 (b) 2:3
(c) 4:1 (d) 6:1

38. A large cube is formed from the material obtained by melting three smaller cubes of 3, 4 and 5 cm side. What is the ratio of the total surface areas of the smaller cubes and the large cube?

(a) 2:1 (b) 3:2
(c) 25:18 (d) 27:20

39. How many small cubes, each of 96 cm^2 surface area, can be formed from the material obtained by melting a larger cube with 384 cm^2 surface area?

(a) 8 (b) 5
(c) 800 (d) 8000

40. A cubical metallic tank whose each edge measures 30 cm, is completely filled with water. If 2,7 litres water is taken out of it, what will be the depth of the remaining water in the tank?

(a) 37 cm (b) 27 cm
(c) 17 cm (d) None of these

41. Find out the weight of a hollow cylindrical lead pipe 28 cm long and $\frac{1}{2}$ cm thick. Its internal diameter is 8 cm. $\left(\text{Weight of 1 } cm^3 \text{ of lead is 11.4 g, } \pi = \frac{22}{7}\right)$

(a) 3.762 Kg (b) 4.562 Kg
(c) 7.462 Kg (d) 6.762 Kg

42. Volume of the cylinder is 1650 m^3. whereas the surface area of its base is $78\frac{4}{7}$ m^2. Find out the height of the cylinder.

(a) 2.1 m (b) 7.5 m
(c) 21 m (d) 14 m

43. 1496 cm^3 of a metal is used to cast a pipe of length 28 cm. If the internal radius of the pipe is 8 cm, then the outer radius of the pipe is:

(a) 7 cm (b) 9 cm
(c) 10 cm (d) 12 cm

44. The base of a of 10 cm high solid cylinder is a semi-circle of radius 7 cm. Its total surface (in cm^2) is $\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) 154 (b) 176
(c) 514 (d) None of these

45. A sphere is melted to form a cylinder whose height is $4\frac{1}{2}$ times its radius; What is the ratio of radii of sphere to the cylinder?

(a) 3:2 (b) 4:3
(c) 3:5 (d) 2:3

46. The radius of a cylinder is the same as that of a sphere. Their volumes are equal. The height of the cylinder is:

(a) 4/3 times its radius. (b)2/3 times its radius.
(c) equal to its radius. (d)equal to its diameter.

47. A 12 m deep well with internal diameter 3.5 m is dug up. The earth from it is spread evenly to form a platform 10.5 m by 8.8 m. Determine the height of the platform.

(a) 2.25 m (b) 3.25 m
(c) 1.25 m (d) 4.25 m

48. The curved surface of a well is 264 sq m and its capacity is 924 m^3. What is the diameter and the depth of the well?

(a) 8 m (b) 9 m
(c) 4.5 m (d) 6 m

49. The sum of the radius of the base and the height of a solid cylinder is 37 m. If the total surface area of the cylinder be 1628 m^2, find out the volume:

(a) 4620 m^3 (b) 4630 m^3
(c) 4520 m^3 (d) 4830 m^3

50. A brick measures 20 cm by 10 cm by $7\frac{1}{2}$ cm. How many bricks will be required for constructing a 25 m long, 2 m high and $\frac{3}{4}$ m thick wall?

(a) 25000 (b) 35000
(c) 20000 (d) 45000

51. The height of a right circular cylinder is 6 m. 3 times the sum of the areas of its two circular faces is twice the area of its curved surface. The radius of the base is:

(a) 4 m (b) 2 m
(c) 6 m (d) 1.5 m

52. The radius of the cylinder is made twice as large. How should the height be changed so that the volume remains the same?

(a) $\frac{1}{2}$ × height of two cylinders

(b) $\frac{1}{4}$ × height of original cylinder

(c) $\frac{1}{4}\ \pi r^2$

(d) None of these

53. A right cylinder and a right circular cone have the same radius and the same volume. The ratio of the height of the cylinder to that of the cone is:

(a) 3:5 (b) 2:5
(c) 3:1 (d) 1:3

54. 2 cans have the same height equal to 21 m. One can is cylindrical, the diameter of whose base is 10 cm. The other can has square base of side 10 cm. What is the difference in between their capacities?

(a) 350 cm^2 (b) 450 cm^3
(c) 250 cm^2 (d) None of these

55. A roller is 120 cm long and has diameter 84 cm. If it takes 500 complete revolutions to level a play ground, then determine the cost of levelling at the rate of 30 paise per m^2. $\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) ₹475.40 (b) ₹375.45
(c) ₹375.20 (d) ₹475.20

56. The circumference of one end of a frustum of a right circular cone is 48 cm and of the other end is 34 cm. If the height of the frustrum is 10 cm, its volume (in cm^3) is:

(a) 5400 (b) 1350
(c) 2700 (d) 4050

57. Find out the amount of concrete required to erect a concrete pillar whose circular base will have a perimeter 8.8 m and whose curved surface is 17.6 m. $\left(\pi = \frac{22}{7}\right)$

(a) $12\frac{4}{25}\ m^3$ (b) $12\frac{3}{25}\ m^3$

(c) $12\frac{1}{2}\ m^3$ (d) $12\frac{8}{25}\ m^3$

58. Sum of the length, width and depth of a cuboid is s and its diagonal is d. Its surface area is:

(a) s^2 (b) d^2
(c) $s^2 - d^2$ (b) $s^2 + d^2$

59. A cylindrical tower is 5 m in diameter and 14 m high. The cost of white washing its curved surface at 50 paise per m^2 is:

(a) ₹90 (b) ₹97
(c) ₹100 (d) ₹110

60. A solid piece of iron of dimensions 49 × 33 × 24 cm is moulded into a sphere, The radius of the sphere is:

(a) 35 cm (b) 21 cm
(c) 29 cm (d) None of these

61. How many coins, 2 mm thick and 1.5 cm in diameter, should be melted in order to form a right circular cylinder its base diameter 6 cm and height 8 cm?

(a) 640 (b) 540
(c) 740 (d) 840

62. A hemisphere is made of lead. Its of radius is 6 cm; It cast into a right circular cone of 75 cm height. The radius of the base of the cone is:

(a) 1.4 cm (b) 2.4 cm
(c) 1.6 cm (d) 3.2 cm

63. A solid cylinder has a total surface area of 231 cm^2. Its curved surface area is (2/3) of the total surface area. Find out the volume of the cylinder.

(a) 270 cm^3 (b) 269.5 cm^3
(c) 256.5 cm^3 (d) 289.5 cm^3

64. It is required to design a circular pipe such that water flowing through it at a speed of 7 m per min fills a tank of capacity 440 cubic m in 10 min. The inner radius of the pipe should be:

(a) 2 m (b) $\sqrt{2}$ m
(c) $\frac{1}{2}$ m (d) $\frac{1}{\sqrt{2}}$ m

65. From a solid right circular cylinder with height 10 cm and radius of the base 6 cm; a right circular cone of the same height and base is removed. The volume (in cm^3) of the remaining solid is:

(a) 377 (b) 754.3
(c) 1131 (d) None of these

66. The radii of two cylinders are in the ratio 2:3. The ratio their height is 5:3. The ratio of their volume is:

(a) 20:27 (b) 10:9
(c) 18:13 (d) 9:20

67. The capacity of a tank, in the form of a cylinder, is 6160 m^3. If the diameter of its base is 28 m, find out the cost of painting its inner curved surface at the rate of ₹2.8 per m^2. $\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) 2464 (b) 2664
(c) 3064 (d) 2864

68. The sum of the radius of the base and the height of a solid cylinder is 37 m. If the total surface area of the solid cylinder is 1628 m^2, then the circumference of its base and the volume of the cylinder are:

(a) 68 m; 7875 m^3 (b) 52 m; 5825 m^3
(c) 44 m; 4620 m^3 (d) 30 m; 3859 m^3

69. A rectangular piece of paper is 22 cm long and 10 cm wide. A cylinder is formed by rolling the paper along its length. The volume of the cylinder is:

(a) 225π cm^3 (b) 385 cm^3
(c) 25π cm^3 (d) None of these

70. The ratio of total surface area to lateral surface area of a cylinder whose radius is 80 cm and height 20 cm, is:

(a) 2:1 (b) 3:1
(c) 4:1 (d) 5:1

71. A right cylindrical vessel is full with water. How many right cones having the same diameter and height as those of right cylinder will be needed to store that water?

(a) 2 (b) 3
(c) 4 (d) 5

72. A cylindrical bucket is 72 cm high and 28 cm in diameter and is full of water. This water is emptied in a rectangular tank whose length and breadth are 66 cm and 28 cm, respectively. What will be the height of the water level in the tank?

(a) 36 cm (b) 48 cm
(c) 24 cm (d) 22 cm

73. A cylindrical iron rod is 70 cm long. The diameter of its end portion is 2 cm. What is its weight, reckoning a cm^3 of iron to weigh 10 grams?

(a) 4 Kg (b) 4.2 Kg
(c) 2.2 Kg (d) Data inadequate

74. If the radius of a cylinder is doubled and the height is halved, then what would be ratio between the new curved surface area and the previous curved surface area of the cylinder:

(a) 1:1 (b) 2:1
(c) 3:2 (d) 2:3

75. A cylindrical jar of diameter 24 cm contains water to a height of 30 cm. A spherical steel ball is dropped into the jar and the level of the water rises by 67.5 mm. The diameter of the ball is:

(a) 16 cm (b) 15 cm
(c) 20 cm (d) 18 cm

76. The material of a solid cone is converted into the shape of solid cylinder of equal radius. If the height of the cylinder is 5 cm, then what is the height of the cone?

(a) 25 cm (b) 15 cm
(c) 20 cm (d) 10 cm

77. The volume of solid cylinder whose diameter of the base is 14 mm and length 25 mm is 3850 mm^3. If length of the cylinder is doubled, but the diameter is halved, then what will be the volume of the resulting cylinder?

(a) 1172 mm^3 (b) 1925 mm^3
(c) 3850 mm^3 (d) 7700 mm^3
(e) None of these

78. A monument has 50 cylindrical pillars each of diameter 50 cm and height 4 m. What will be the labour charges for cleaning these pillars at the rate of 50 paise per m^2? (Use $\pi = 3.14$):

(a) ₹237 (b) ₹257
(c) ₹157 (d) ₹353

79. The radius of a cylinder is made twice large. How should the height be changed, so that its volume remains unchanged?

(a) $\frac{1}{4}$ of original (b) $\frac{1}{3}$ of original
(c) $\frac{1}{2}$ of original (d) $\frac{1}{8}$ of original

80. A spherical ball of lead, 3 cm in diameter, is melted and re-cast into three spherical balls. The diameter of 2 of these are 1.5 cm and 2 cm, respectively. The diameter of the third ball is:

(a) 2.66 cm (b) 2.5 cm
(c) 3 cm (d) 3.5 cm

81. A cone and a cylinder having the same area of the base have also the same area of curved surfaces. If the height of cylinder be 2 m, find out the slant height of the cone:

(a) 3 m (b) 3.5 m
(c) 4.5 m (d) 4 m

82. The radii of a cylinder and a cone are equal. If the height of the cylinder is equal to the slant height of the cone then the ratio of the curved surfaces of the cylinder and the cone is:

(a) 1:1 (b) 2:1
(c) 3:1 (d) 4:1

83. From a cubical block of wood of side 1 m, a cylinder of the largest possible volume is curved out. The volume (in m^3) of the remaining wood is:

(a) $\frac{3}{14}$ (b) $\frac{5}{14}$
(c) $\frac{1}{2}$ (d) $\frac{2}{7}$

84. The radius of the base of a solid cylinder is r cm and its height is 3 cm. It is re-casted into a cone of same radius, the height of the cone will be:

(a) 3 cm (b) 6 cm
(c) 9 cm (d) 27 cm

85. 2 cm of rain has fallen on a Km^2 of land. Assuming that 50% of the raindrops could have been collected and contained in a pool having a 100 m × 10 m base, by what level would the water level in the pool have increased?

(a) 15 m (b) 20 m
(c) 10 m (d) 25 m

86. The perpendicular height of a conical tent is $4\frac{2}{3}$ m and the diameter of its base is 6 m. If 11 persons can sleep in this tent, find how many average cu m of air each person gets?

(a) 2 cu m (b) 4 cu m
(c) 6 cu m (d) 8 cu m

87. The circumference of the base of a 9 m high conical tent is 44 m. The volume of the air contained in it is:

(a) 462 m^3 (b) 452 m^3
(c) 472 m^3 (d) 512 m^3

88. A conical vessel of base radius 2 cm and height 3 cm is filled with kerosene. This liquid leaks through a hole in the bottom and collects in a cylindrical jar of radius 2 cm. The kerosene level in the jar is:

(a) 1.5 cm (b) π cm
(c) 1 cm (d) 3 cm

89. A hollow cylinder of height 3 cm, is re-casted into a solid cylinder. If the external and internal radii of the hollow cylinder are 4.3 cm and 1.1 cm, respectively. What will be the radius of the solid cylinder?

(a) 2.8 cm (b) 2.4 cm
(c) 3.2 cm (d) 4.8 cm

90. A solid consists of a circular cylinder with an exact fitting right circular cone placed on the top. The height of the cone is h. If the total volume of the solid is three times the volume of the cone, then the height of the cylinder is:

(a) $2h$ (b) $4h$

(c) $\frac{2h}{3}$ (d) $\frac{3h}{3}$

91. A well of 11.2 m diameter is dug 8 m deep. The earth taken out has been spread all around it to a width of 7 cm to form a circular embankment. Find out the height of the embankment.

(a) 304.8 m^2 (b) 400.4 m^2
(c) 408.4 m^2 (d) 412.4 m^2

92. The curved surface of a circular cylinder of height h and the slant surface of the cone of slant height '$2h$' having the same circular base are in the ratio of:

(a) 1:1 (b) 1:2
(c) 3:2 (d) 1:3

93. The material of a cone is converted into the shape of a cylinder of equal radius. If the height of the cylinder is 5 cm, the height of the cone is:

(a) 10 cm (b) 15 cm
(c) 18 cm (d) 24 cm

94. A right circular cone is exactly fitted inside a cube in such a way that the edges of the base of the cone are touching the edges of one of the faces of the cube and the vertex is on the opposite face of the cube. If the volume of the cube is 343 c.c, then what approximately is the volume of the cone?

(a) 90 c.c. (b) 75 c.c.
(c) 80 c.c. (d) 85 c.c.

95. A solid cone is 25 cm high and the radius of its base is 50 cm. It is melted and re-cast into a solid sphere. Determine the surface area of the sphere.

(a) 8757.28 cm^2 (b) 5877.42 cm^2
(c) 7857.14 cm^2 (d) None of these

96. The radius and height of right circular cone are in the ratio 5:12. If its volume is 314 $\frac{3}{7}$ m^3. Find out the radius of the cone.

(a) 5 m (b) 8 m
(c) 12 m (d) 6 m

97. A cone, a hemisphere and a cylinder stand on equal bases and have the same height. The ratio of their volumes are:

(a) 1:2:2 (b) 1:2:3
(c) 1:2:4 (d) 2:3:4

98. Find out the length of the canvas 2 m in width required to make a conical tent 12 m in diameter and 6.3 m in slant height:

(a) 118.8 m (b) 62.4 m
(c) 59.4 m (d) 112.4 m

99. The radius of a cylinder is doubled and the height is halved, what is the ratio between the new volume and the previous volume?

(a) 3:1 (b) 2:3
(c) 2:1 (d) 1:3

100. A circus tent is cylindrical to a height of 3 m and conical above it. If its diameter is 105 m and slant height of the conical portion is 53 m, then calculate the length of the canvas 5 m wide to make the tent.

(a) 1857 m (b) 1647 m
(c) 1947 m (d) 1847 m

101. If base radius of a cone is increased by 20% and its slant height is doubled, then by how much per cent will the area of its curved surface be increased?

(a) 140% (b) 160%
(c) 130% (d) 180%

102. The radius of the base of conical tent is 5 cm. If the tent is 12 m high, then area of the canvas required in making the tent is:

(a) 60 π m^2 (b) 300 π m^2
(c) 90 π m^2 (d) None of these

103. A cone of height 7 cm and base radius 3 cm is carved from a rectangular block of wood 10 cm × 5 cm × 2 cm. The percentage % wood wasted is:

(a) 34% (b) 46%
(c) 54% (d) 66%

104. The diameter and slant height of a conical tomb are 28 m and 50 m, respectively. The cost of white washing its curved surface at the rate of 80 paise per m^2 is:

(a) ₹2640 (b) ₹1760
(c) ₹264 (d) ₹176
(e) None of these

105. A rectangular sheet of area 264 cm^2 and width 11 cm is rolled along its breadth to make a hollow cylinder. The volume of the cylinder is:

(a) 231 c.c. (b) 230 c.c.
(c) 235 c.c. (d) 234 c.c.

106. A cylinder and a cone have their heights in the ratio 2:3 and the radii of their bases in the ratio 3:4. Find out the ratio of their volumes.

(a) 1:9 (b) 2:9
(c) 9:8 (d) 1:8

107. If the height of a cone is doubled, then its volume is increased by:

(a) 100% (b) 200%
(c) 300% (d) 400%

108. 3 cubes of side 3, 4 and 5, respectively, are melted to form into new cube. The side of the new cube is:

(a) 5 cm (b) 6 cm
(c) 6.5 cm (d) 7 cm

109. The height and base radius of a cone are each increased by 100%. The volume of the cone now becomes:

(a) double the original.
(b) 4 times the original.
(c) 3 times the original.
(d) 8 times the original.

110. If the radius of a sphere is doubled, then its volume is increased by:

(a) 100% (b) 200%
(c) 700% (d) 800%

111. The diameter of a sphere is 6 cm. It is melted and drawn into a wire of diameter 0.2 cm. Find out the length of the wire.

(a) 24 m (b) 28 m
(c) 36 m (d) 32 m

112. A cone-shaped circular tent is 9 m high and the circumference of its circular base is 44 m. How much air is contained in the tent? $\left(\text{Use } \pi = \frac{22}{7}\right)$

(a) 362 m^3 (b) 462 m^3
(c) 562 m^3 (d) 662 m^3

113. If the radius of a sphere is doubled, then its surface area is increased by:

(a) 100% (b) 200%
(c) 300% (d) 50%

114. The height of a cylinder is decreased by 8%, keeping its radius unchanged. What is the percentage change in its volume?

(a) 8% increase (b) 12% decrease
(c) 8% decrease (d) None of these

115. The radius of a cylinder is increased by 20%, keeping its height unchanged. What is the percentage increase in its volume?

(a) 33% (b) 44%
(c) 22% (d) None of these

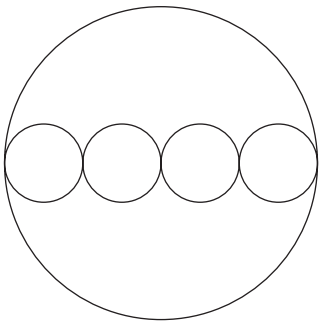
EXERCISE-2
(BASED ON MEMORY)

1. The radius of base of solid cone is 9 cm. and its height is 21 cm. It cut into 3 parts by two cuts which are parallel to its base. The cuts are at height of 7 cm. and 14 cm. from the base respectively. What is the ratio of curved surface area of top, middle and bottom parts respectively?

(a) 1: 4: 8 (b) 1: 3: 5
(c) 1: 3: 9 (d) 1: 6: 12

[SSC CGL Tier-II CBE, 2018]

2. A solid cylinder has radius of base 14 cm. and height 15 cm. identical cylinders are cut from each base as shown in the given figure. Height of small cylinder is 5 cm. what is the total surface area (In cm^2) of the remaining part?



(a) 3740 (b) 3432
(c) 3124 (d)42818

[SSC CGL Tier-II CBE, 2018]

3. Find the curved surface area (in cm^2) of a hemisphere of diameter 28 cm.

(a) 1152 (b) 1024
(c) 956 (d) 1232

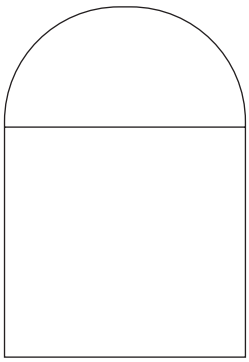
[SSC CHSL (10+2) Tier-I CBE, 2018]

4. The ratio of curved surface area and volume of a cylinder is 1:7. The ratio of total surface area and volume is 187:770. What is the respective ratio of its base radius and height?

(a) 5:8 (b) 4:9
(c) 3:7 (d) 7:10

[SSC CGL Tier-II CBE, 2018]

5. A hemisphere is kept on top of a cube. Its front view is shown in the given figure. The total height of the figure is 221 cm. The ratio of curved surface area of hemisphere and total surface area of cobe is 11:42. What is the total volume (in cm) of figure?



(a) 3318.33 (b) 3462.67
(c) 3154.67 (d) 3248.33

[SSC CGL Tier-II CBE, 2018]

6. 10 identical solid spherical balls of radius 3 cm. are melted to form a single sphere. In this process 20% of solid is wasted. What is the radius (in cm) of the biggest sphere?

(a) 24 (b) 12
(c) 8 (d) 6

[SSC CGL Tier-II CBE, 2018]

7. A right circular cylinder has height as 18 cm and radius as 7 cm. The cylinder is cut in the three equal parts (by 2 cuts parallel to base). What is the percentage increase in total surfacearea?

(a) 62 (b) 56
(c) 48 (d) 52

[SSC CGL Tier-II CBE, 2018]

8. Radius of base of a hollow cone is 8 cm. and its height is 15 cm. A sphere of largest radius is put inside the cone. What is the ration of base of cone to the radius of sphere?

(a) 5:3 (b) 4:1
(c) 2:1 (d) 7:3

[SSC CGL Tier-II CBE, 2018]

9. The Total surface area of a hemisphere is 41.58 sq. cm. Find its curved surface area.

(a) 27.72 sq. cm. (b) 55.44 sq. cm.
(c) 9.24 sq. cm. (d) 13.86 sq. cm.

[SSC CGL Tier-I CBE, 2017]

10. What is the curved surface area (In cm^2) of a cylinder having radius of base as 14 cm and height as 10 cm?

(a) 440 (b) 880
(c) 220 (d) 1320

[SSC CAPFs ASI & Delhi Police SI, 2017]

11. The perimeter of base of a right circular cone is 132 cm. If the height of the cone is 72 cm. then what is the total surface area (In cm^2) of the cone?

(a) 6600 (b) 6336
(c) 4224 (d) 5784

[SSC CAPFs ASI & Delhi Police SI, 2017]

12. A solid right circular cone of radius 4 cm and height 7 cm is put inside a cylindrical vessel of radius 5 cm and height 8 cm. How much water in cube cm will be required to fill the cylindrical vessel completely?

(a) 1022.48 cubic cm (b) 1533.72 cubic cm
(c) 511.24 cubic cm (d) 255.62 cubic cm

[SSC CGL Tier-I CBE, 2017]

13. The cross section of a canal is in the shape of an isosceles trapezium which is 4 metre wide art the bottom and 5 metre wide at the top. If the depth of the canal is 2 metre and it is 120 metre long, what is the maximum capacity of this canal?

(a) 2160 cubic metre (b) 3240 cubic metre
(c) 4320 cubic metre (d) 1080 cubic metre

[SSC CHSL (10+2) Tier-I CBE, 2017]

14. The curved surface area of a hemisphere is 27.72 square cm and volume is 19.404 cubic cm, find its radius.

(a) 4.2 cm. (b) 2.1 cm.
(c) 0.7 cm. (d) 1.05 cm.

[SSC CGL Tier-I CBE, 2017]

15. What is the volume (in cm^3) of a right pyramid of height 12 cm and having a square base whose diagonal is $6\sqrt{2}$ cm?

(a) 864 (b) 432
(c) 144 (d) 288

[SSC CAPFs ASI & Delhi Police SI, 2017]

16. A solid cone of height 24 cm and having radius of base 8 cm is melted to form a solid cylinder of radius 6 cm and height 6cm. In the whole process what percent of material is wasted?

(a) 48.5 (b) 37.5
(c) 57.8 (d) 64

[SSC CGL Tier-I CBE, 2017]

17. A solid cylinder having radius of base as 7 cm. and length as 20 cm. is bisected from its height to get two identical cylinders. What will be the percentage increase in the total surface area?

(a) 29.78 (b) 25.93
(c) 27.62 (d) 32.83

[SSC CAPFs ASI & Delhi Police SI, 2017]

18. Three sphereical balls of radius 2 cm, 4 cm and 6 cm are melted to form a new apherical ball. In this process there is a loss of 25% of the material. What is the radius (in cm) of the new ball?

(a) 6 (b) 8
(c) 12 (d) 16

[SSC CAPFs ASI & Delhi Police SI, 2017]

19. A solid cone of height 36 cm and radius of base 9 cm is melted to form a solid cylinder of radius 9 cm and height 9 cm. What percent of material is wasted in this process?

(a) 25 (b) 0
(c) 5 (d) 10

[SSC CGL Tier-I CBE, 2017]

20. The radius of a wheel is 3.5 cm. What is the distance (in cm) travelled by the wheel in 20 revolutions?

(a) 220 (b) 440
(c) 880 (d) 1320

[SSC CAPFs ASI & Delhi Police SI, 2017]

21. What is the length (in metre) of the longest rod that can be placed in a room which is 2 metre long 2 metre broad and 6 metre high?

(a) 8 (b) $2\sqrt{11}$
(c) $3\sqrt{11}$ (d) 10

[SSC CAPFs ASI & Delhi Police SI, 2017]

22. The radius of a wheel is 21 cm. What is the distance (in cm) travelled by the wheel in 10 revolutions?

(a) 660 (b) 1320
(c) 1980 (d) 2640

[SSC CAPFs ASI & Delhi Police SI, 2017]

23. A right circular cylindrical tunnel of diameter 5m and length 10m is to be constructed from a sheet of iron. The area of iron sheet required will be:

(a) 52π (b) 50π
(c) 51π (d) 49π

[SSC CAPFs (CPO) SI & ASI, Online, 2016]

24. A rectangular water tank is 80 metre x 40 metre. Water flows into it through a pipe of 40 sq. cm. at the opening at a speed of 10 km/hr. The water level will rise in the tank in half an hour by

(a) $\frac{3}{2}$ cm. (b) $\frac{4}{9}$ cm.

(c) $\frac{5}{9}$ cm. (d) $\frac{5}{8}$ cm.

[SSC CGL Tier-II, 2016]

25. A right circular cylinder having diameter 21 cm and height 38 cm is full of ice cream. The ice cream is to be filled in cones of height 12 cm and diameter 7 cm having a hemispherical shape on the top. The number of such cones to be filled with ice cream is

(a) 54 (b) 44
(c) 36 (d) 24

[SSC CGL Tier-II, 2016]

26. A spherical aquarium can accommodate 11 fishes, and each fish requires 1.54 cu. Metre of water. What is the volume of the aquarium?

(a) 11.14 cu. metre (b) 16.94 cu. Metre
(c) 10.25 cu. Metre (d) 17.84 cu. Metre

[SSC CPO, 2016]

27. A prism with a right triangular base is 25 cm high. If the shorter sides of the triangle are in the ration of 1:2 and the volume of the prism is 100 cm^3 what is the length of the longest side of the triangle?

(a) $\sqrt{5}$ cm (b) $2\sqrt{5}$ cm
(c) $5\sqrt{2}$ cm (d) 5 cm

[SSC CAPFs (CPO) SI & ASI, Delhi Police, 2016]

28. A hollow hemispherical bowl is made of silver with its outer radius 8 cm and inner radius 4 cm respectively. The bowl is melted to form a solid right circular cone of radius 8 cm. The height of the cone formed is

(a) 7 cm (b) 9 cm
(c) 12 cm (d) 14 cm

[SSC CGL Tier-I (CBE), 2016]

29. A solid sphere and a solid hemisphere have the same total surface area. The ratio of their volumes is $\left(\text{Taker}, \pi = \frac{22}{7}\right)$

(a) $3\sqrt{3} : 4$ (b) $4 : 3\sqrt{3}$
(c) $3 : 4\sqrt{3}$ (d) $1 : 12\sqrt{3}$

[SSC CGL Tier-II (CBE), 2016]

30. Three solid spheres have their radii r_1, r_2 and r_3. The spheres are melted to form a solid spheres of bigger radius, Then the radius of the new spheres is:

(a) $(r_1 + r_2 + r_3)$ (b) $(r_1^2 + r_2^2 + r_3^2)^{\frac{1}{2}}$
(c) $(r_1^3 + r_2^3 + r_3^3)^{\frac{1}{3}}$ (d) $(r_1^4 + r_2^4 + r_3^4)^{\frac{1}{4}}$

[SSC CGL Tier-I (CBE), 2016]

31. An inverted conical shaped vessel is filled with water to its brim. The height of the vessel is 8 cm and radius of the open end is 5 cm. When a few solid spherical metallic balls each of radius $\frac{1}{2}$ cm are dropped in the vessel.25% water is overflowed. The number of balls is:

(a) 100 (b) 400
(c) 200 (d) 150

[SSC CGL Tier-I (CBE), 2016]

32. The respective ratio of curved surface area and total surface area of a cylinder is 4 : 5. If the curved surface area of the cylinder is 1232 cm^2, what is the height? (in cm)

(a) 14 cm (b) 28 cm
(c) 7 cm (d) 56 cm
(e) 24 cm

[IBPS, 2015]

33. A plane divides a right circular cone into two parts of equal volume. If the plane is parallel to the base, then the ratio, in which the height of the cone is divided, is

(a) $1 : \sqrt[3]{2} - 1$ (b) $1 : \sqrt{2}$
(c) $1 : \sqrt[3]{2}$ (d) $1 : \sqrt[3]{2} + 1$

[SSC, 2015]

34. A right prism has a triangular base whose sides are 13 cm, 20 cm and 21 cm. If the altitude of the prism is 9 cm, then its volume is

(a) 1413 cm^3 (b) 1314 cm^3
(c) 1143 cm^3 (d) 1134 cm^3

[SSC, 2015]

35. A cylinder with base radius 8 cm and height 2 cm is melted to form a cone of height 6 cm. The radius of the cone will be

(a) 5 cm (b) 4 cm
(c) 8 cm (d) 6 cm

[SSC, 2015]

36. The portion of a ditch 48 m long, 16.5 m wide and 4 m deep that can be filled with stones and earth available during excavation of a tunnel, cylindrical in shape, of diameter 4 m and length 56 m is $\left(\text{Take}\pi = \frac{22}{7}\right)$

(a) $\frac{1}{2}$ Part (b) $\frac{1}{4}$ Part
(c) $\frac{2}{9}$ Part (d) $\frac{1}{9}$ Part

[SSC, 2015]

37. The numerical values of the volume and the area of the lateral surface of a right circular cone are equal. If the height of the cone be h and radius be *r*, the value of $\frac{1}{h^2} + \frac{1}{r^2}$ is

(a) $\frac{9}{1}$ (b) $\frac{1}{9}$
(c) $\frac{3}{1}$ (d) $\frac{1}{3}$

[SSC, 2015]

38. The radii of two solid iron spheres are 1 cm and 6 cm respectively. A hollow sphere is made by melting the two spheres. If the external radius of the hollow sphere is 9 cm, then its thickness (in cm) is

(a) 1.5 (b) 2
(c) 0.5 (d) 1

[SSC, 2015]

39. If a hemisphere is melted and four spheres of equal volume are made, the radius of each sphere will be equal to

(a) 1/2 of the radius of the hemisphere
(b) $1/6^{th}$ of the radius of the hemisphere
(c) radius of the hemisphere
(d) $1/4^{th}$ of the radius of the hemisphere

[SSC, 2015]

40. There is a wooden sphere of radius 6 $\sqrt{3}$ cm. The surface area of the largest possible cube cut out from the sphere will be

(a) 464 $\sqrt{3}$ cm^2 (b) 864 cm^2

(c) 462 cm^2 (d) 646 $\sqrt{3}$ cm^2

[SSC, 2015]

41. If 64 buckets of water are removed from a cubical shaped water tank completely filled with water, 1/3 of the tank remains filled with water. The length of each side of the tank is 1.2 m. Assuming that all buckets are of the same measure, then the volume (in litres) of water contained by each bucket is

(a) 15 (b) 18

(c) 16 (d) 12

[SSC, 2015]

42. Base of a right pyramid is a square of side 10 cm. If the height of the pyramid is 12 cm, then its total surface area is

(a) 460 cm^2 (b) 260 cm^2

(c) 360 cm^2 (d) 400 cm^2

[SSC, 2015]

43. The length of canvas 75 cm wide required to build a conical tent of height 14 m and the floor area 346.5 m^2 is:

(a) 770 m (b) 490 m

(c) 860 m (d) 665 m

[SSC, 2015]

44. If the area of the base, height and volume of a right prism be $\left(3\sqrt{3}/2\right)P^2 cm^2, 100\sqrt{3}$ and 7200 cm^3 respectively, then the value of P will be?

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{3}{2}$

(c) $\sqrt{3}$ (d) 4

[SSC, 2015]

45. If the volume of a sphere is numerically equal to its surface area then its diameter is:

(a) 4 cm (b) 2 cm

(c) 3 cm (d) 6 cm

[SSC, 2015]

46. 5 persons will live in a tent. If each person requires $16m^2$ of floor area and $100m^3$ space for air then the height of the cone of smallest size to accommodate these persons would be?

(a) 18.75 m (b) 16 m.

(c) 10.25 m (d) 20 m

[SSC, 2015]

47. The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. If its volume be $\frac{1}{27}$th of the volume of the given cone, at what height above the base is the section made?

(a) 19 cm (b) 20 cm

(c) 12 cm (d) 15 cm

[SSC Examination, 2014]

48. If the surface area of a sphere is 346.5 cm^2, then its radius is $\left[\text{taking } \pi = \frac{22}{7}\right]$

(a) 7 cm (b) 3.25 cm

(c) 5.25 cm (d) 9 cm

[SSC Examination, 2014]

49. The height of the right pyramid whose area of the base is 30 m^2 and volume is 500 m^3, is:

(a) 50 m (b) 60 m

(c) 40 m (d) 20 m

[SSC Examination, 2014]

50. The base of a prism is a right-angled triangle with 2 sides 5 cm and 12 cm. The height of the prism is 10 cm. The total surface area of the prism is :

(a) 360 cm^2 (b) 300 cm^2

(c) 330 cm^2 (d) 325 cm^2

[SSC Examination, 2014]

51. The base of a right prism is an equilateral triangle. If the lateral surface area and volume is 120 cm^2 and $40\sqrt{3}$ cm^3, respectively, then the side of base of the prism is:

(a) 4 cm (b) 5 cm

(c) 7 cm (d) 40 cm

[SSC Examination, 2014]

52. A ball of lead, 4 cm in diameter, is covered with gold. If the volume of the gold and lead are equal, then the thickness of gold is approximately [given $\sqrt[3]{2} = 1.259$]

(a) 5.038 cm (b) 5.190 cm

(c) 1.038 cm (d) 0.518 cm

[SSC Examination, 2014]

53. A large solid sphere is melted and moulded to form identical right circular cones with base radius and height same as the radius of the sphere. One of

these cones is melted and moulded to form a smaller solid sphere. Then the ratio of the surface area of the smaller to the surface area of the larger sphere is:

(a) $1:3^{\frac{4}{3}}$ (b) $1:2^{\frac{3}{2}}$

(c) $1:3^{\frac{2}{3}}$ (d) $1:2^{\frac{4}{3}}$

[SSC Examination, 2014]

54. A conical cup is filled with ice cream. The ice cream forms a hemispherical shape on its open top. The height of the hemispherical part is 7 cm. The radius of the hemispherical part equals the height of the cone. Then the volume of the ice cream is $\left[\pi = \frac{22}{7}\right]$

(a) 1078 m^3 (b) 1708 m^3

(c) 7108 m^3 (d) 7180 m^3

[SSC Examination, 2014]

55. If each side of a cube is increased by 10%, the volume of the cube will increase by:

(a) 30% (b) 10%

(c) 33.1% (d) 25%

[SSC Examination, 2014]

56. A rectangular plot, 36 m long and 28 m broad, has two concrete roads 5 m wide running in the middle of the park,one parallel to the length and the other parallel to the breadth. What would be the total cost of gravelling the plot, excluding the area covered by the roads, at ₹3.60 per m^2?

(a) ₹2772.20 (b) ₹2466.60

(c) ₹2654.40 (d) ₹2332.60

(e) ₹2566.80

[IBPS PO/MT Examination, 2014]

57. A sphere is cut into two hemispheres. One of them is used as bowl. It takes 8 bowlfuls of this to fill a conical vessel of height 12 cm and radius 6 cm. The radius of the sphere (in centimeter) will be

(a) 3 (b) 2

(c) 4 (d) 6

[SSC, 2014]

58. The base of a right prism is a quadrilateral ABCD. Given that AB = 9 cm, BC = 14 cm, CD = 13 cm, DA = 12 cm and $\angle DAB = 90°$. If the volume of the prism be 2070 cm^3, then the area of the lateral surface is

(a) 720 cm^2 (b) 810 cm^2

(c) 1260 cm^2 (d) 2070 cm^2

[SSC, 2014]

59. The volumes of a right circular cylinder and a sphere are equal. The radius of the cylinder and the diameter of the sphere are equal. The ratio of height and radius of the cylinder is

(a) 3 : 1 (b) 1 : 3

(c) 6 : 1 (d) 1 : 6

[SSC, 2014]

60. A rectangular tin sheet is 12 cm long and 5 cm broad. It is rolled along its length to form a cylinder by making the opposite edges just to touch each other. Then the volume of the cylinder is

(a) $\frac{100}{\pi}\text{cm}^3$ (b) $\frac{60}{\pi}\text{cm}^3$

(c) $\frac{180}{\pi}\text{cm}^3$ (d) $\frac{120}{\pi}\text{cm}^3$

[SSC, 2014]

61. The area of the iron sheet required to prepare a cone 24 cm high with base radius 7 cm is $(Take \pi = \frac{22}{7})$

(a) 704 cm^2 (b) 408 cm^2

(c) 708 cm^2 (d) 804 cm^2

[SSC, 2014]

62. Some bricks are arranged in an area measuring 20 cu. m. If the length, breadth and height of each brick is 25 cm, 12.5 cm and 8 cm respectively, then in that pile the number of bricks are (suppose there is no gap in between two bricks)

(a) 4,000 (b) 10,000

(c) 6,000 (d) 8,000

[SSC, 2014]

63. A right circular cone is 3.6 cm high and radius of its base is 1.6 cm. It is melted and recast into a right circular cone with radius of its base as 1.2 cm. Then the height of the cone (in cm) is:

(a) 3.6 (b) 4.8

(c) 6.4 (d) 7.2

[SSC Examination, 2013]

64. If h, c, v are respectively the height, curved surface area and volume of a right circular cone, then the value of $3\pi vh^3 - c^2h^2 + 9v^2$ is:

(a) 2 (b) −1

(c) 1 (d) 0

[SSC Examination, 2013]

65. The volume of a conical tent is 1232 cu m and the area of its base is 154 m^2. Find the length of the canvas required to build the tent, if the canvas is 2 m in width. $\left(\text{Take } \pi = \frac{22}{7}\right)$

(a) 270 m (b)272 m
(c) 276 m (d)275 m

[SSC Examination, 2013]

66. Assume that a drop of water is spherical and its diameter is $\frac{1}{10}$ of a centimetre. A conical glass has a height equal to the diameter of its rim. If 32,000 drops of water fill the glass completely, then the height of the glass (in cm) is:

(a) 1 (b) 2
(c) 3 (d) 4

[SSC Examination, 2013]

67. The total number of spherical bullets, each of diameter 5 decimetre, that can be made by utilizing the maximum of a rectangular block of lead with 11 m length, 10 m breadth and 5 m width is (assume that π>3):

(a) Equal to 8800 (b) Less than 8800
(c) Equal to 8400 (d) Greater than 9000

[SSC Examination, 2013]

68. A rectangular block of metal has dimensions 21 cm, 77 cm and 24 cm. The block has been melted into a sphere. The radius of the sphere is $\left(\text{Take } \pi = \frac{22}{7}\right)$

(a) 21 cm (b) 7 cm
(c) 14 cm (d) 28 cm

[SSC Examination, 2013]

69. If a right circular cone of height 24 cm has a volume of 1232 cm^3, then the area (in cm^2) of curved surface is:

(a) 550 (b) 704
(c) 924 (d) 1254

[SSC Examination, 2013]

70. If each edge of a cube is increased by 50%, the percentage increase in surface area is:

(a) 125% (b) 50%
(c) 100% (d) 75%

[SSC Examination, 2013]

71. If the surface areas of two spheres are in the ratio 4:9, then the ratio of their volumes will be:

(a) 4:9 (b) 16:27
(c) 8:27 (d) 16:9

[SSC Examination, 2013]

72. If each edge of a cube is increased by 50%, the percentage increase in its surface area is:

(a) 150% (b) 75%
(c) 100% (d) 125%

[SSC Assistant Grade III, 2013]

73. The diameter of a copper sphere is 18 cm. The sphere is melted and is drawn into a long wire of uniform circular cross-section. If the length of the wire is 108 m, the diameter of the wire is:

(a) 1 cm (b) 0.9 cm
(c) 0.3 cm (d) 0.6 cm

[SSC Assistant Grade III, 2013]

74. A semicircular sheet of metal of diameter 28 cm is bent into an open conical cup. The capacity of the cup $\left(\text{taking } \pi = \frac{22}{7}\right)$ is:

(a) 624.26 cm^3 (b) 622.36 cm^3
(c) 622.56 cm^3 (d) 623.20 cm^3

[SSC Assistant Grade III, 2013]

75. If surface area and volume of a sphere are S and V, respectively, then value of $\frac{S^3}{V^3}$ is:

(a) 36π (b) 9π
(c) 18π (d) 27π

[SSC Assistant Grade III, 2013]

76. The edge of an ice cube is 14 cm. The volume of the largest cylindrical ice cube that can be formed out of it is

(a) 2200 cm^3 (b) 2000 cm^3
(c) 2156 cm^3 (d) 2400 cm^3
(e) None of these

[IBPS PO/MT Examination, 2013]

77. If the side of a cube is increased by 100%, its volume is increased by:

(a) 400% (b) 800%
(c) 200% (d) 100%

[UPPCS Examination, 2012]

78. A solid sphere of radius 1 cm is melted to convert into a wire of length 100 cm. The radius of the wire (using $\sqrt{3} = 1.732$) is:

(a) 0.08 cm (b) 0.09 cm
(c) 0.16 cm (d) 0.11 cm

[SSC Assistant Grade III, 2012]

79. A field is in the form of a rectangle of length 18 m and width 15 m. A pit, 7.5 m long, 6 m broad and 0.8 m deep, is dug in a corner of the field and the earth taken out is evenly spread over the remaining area of the field. The level of the field raised is:

(a) 12 cm (b) 14 cm
(c) 16 cm (d) 18 cm

[SSC Assistant Grade III, 2012]

80. The base of a right pyramid is an equilateral triangle of side 4 cm. The height of the pyramid is half of its slant height. Its volume is:

(a) $\frac{8}{9}\sqrt{2}$ cm^3 (b) $\frac{7}{9}\sqrt{3}$ cm^3

(c) $\frac{8}{9}\sqrt{3}$ cm^3 (d) $\frac{7}{9}\sqrt{2}$ cm^3

[SSC Assistant Grade III, 2012]

81. Water flows in a tank $150\text{ m} \times 100\text{ m}$ at the base, through a pipe whose cross-section is 2 dm by 1.5 dm, at the speed of 15 Km/h. In what time will the water be 3 m deep?

(a) 100 hours (b) 120 hours
(c) 140 hours (d) 150 hours

[SSC Assistant Grade III, 2012]

82. A tent is of the shape of a right circular cylinder upto a height of 3 m and then becomes a right circular cone with maximum height of 13.5 m above the ground. If the radius of the base is 14 m, the cost of painting the inner side of the tent at the rate of ₹2 per m^2 is:

(a) ₹2,050 (b) ₹2,060
(c) ₹2,068 (d) ₹2,080

[SSC Assistant Grade III, 2012]

83. If the diameter of a sphere is decreased by 25%, its curved surface area will be decreased by:

(a) 43.25% (b) 43.50%
(c) 43.75% (d) 44.25%

[SSC Assistant Grade III, 2012]

84. The radius of a cylinder is 10 cm and height is 4 cm. The number of centim that may be added either to the radius or to the height to get the same increase in the volume of the cylinder is:

(a) 5 (b) 4
(c) 25 (d) 16

[SSC Examination, 2012]

85. If a solid cone of volume 27 π cm^3 is kept inside a hollow cylinder whose radius and height are that of the cone, then the volume of water needed to the empty space is:

(a) 3π cm^3 (b) 18π cm^3
(c) 54π cm^3 (d) 81π cm^3

[SSC Examination, 2012]

86. Two cm of rain has fallen on a square Km of land. Assuming that 50% of the raindrops could have been collected and contained in a pool having a 100 m × 10 m base, by what level would he water level in the pool have increased?

(a) 1 Km (b) 10 m
(c) 10 cm (d) 1 m

[SSC Examination, 2012]

87. A cylindrical can whose base horizontal and is of internal radius 3.5 cm contains sufficient water so that when a solid sphere is placed inside, water just covers the sphere. The sphere fits in the can exactly. The depth of water in the can before the sphere was put is:

(a) $\frac{35}{3}$cm (b) $\frac{17}{3}$cm

(c) $\frac{7}{3}$cm (d) $\frac{14}{3}$cm

[SSC Examination, 2012]

88. The height of a circular cylinder is increased 6 times and the base area is decreased to $\frac{1}{9}$ of its value. The factor by which the lateral surface of the cylinder increases is:

(a) 2 (b) $\frac{1}{2}$

(c) $\frac{2}{3}$ (d) $\frac{3}{2}$

[SSC Examination, 2012]

89. The volume of a right circular cone is 1232 cm^3 and its vertical height is 24 cm. Its curved surface area is:

(a) 154 cm^2 (b) 550 cm^2
(c) 604 cm^2 (d) 704 cm^2

[SSC Examination, 2012]

90. The height of a right prism with a square base is 15 cm. If the area of the total surfaces of the prism is 608 cm^2, its volume is:

(a) 910 cm^2 (b) 920 cm^2
(c) 960 cm^2 (d) 980 cm^2

[SSC Examination, 2012]

91. The volume of a solid hemisphere is 19404 cm^3. Its total surface area is:

(a) 4158 cm^3 (b) 2858 cm^3
(c) 1738 cm^3 (d) 2038 cm^3

[SSC Examination, 2012]

Directions (Q. 92–96): Study the following information and answer the questions that follow:
The premises of a bank are to be renovated. The renovation is in terms of flooring. Certain areas are to be floored either with marble or wood. All rooms/halls and pantry are rectangular. The area to be renovated comprises a hall for customer transaction measuring 23 m by 29 m, the branch manager's room measuring 13 m by 17 m, a pantry measuring 14 m by 13 m, a record keeping-cum-server room measuring 21 m by 13 m and locker area measuring 29 m by 21 m. The total area of the bank is 2000 m^2. The cost of wooden flooring is ₹170 per m^2 and the cost of marble flooring is ₹190 per sm^2. The locker area, record keeping-cum-server room and pantry are to be floored with marble. The branch manager's room and hall for customer transaction are to be floored with wood. No other area is to be renovated in terms of flooring.

92. What is the ratio of the total cost of wooden flooring to the total cost of marble flooring?

(a) 1879:2527 (b) 1887:2386
(c) 1887:2527 (d) 1829:2527
(e) 1887:2351

[IBPS PO/MT Examination, 2012]

93. If the 4 walls and ceiling of the branch manager's room (the height of the room is 12 m) are to be painted at the cost of ₹190 per m^2, how much will be the total cost of renovation of the branch manager's room, including the cost of flooring?

(a) ₹1,36,800 (b) ₹2,16,660
(c) ₹1,78,790 (d) ₹2,11,940
(e) None of these

[IBPS PO/MT Examination, 2012]

94. If the remaining area of the bank is to be carpeted at the rate of ₹110 per m^2, how much will be the increment in the total cost of renovation of bank premises?

(a) ₹5,820 (b) ₹4,848
(c) ₹3,689 (d) ₹6,690
(e) None of these

[IBPS PO/MT Examination, 2012]

95. What is the percentage area of the bank that is not to be renovated?

(a) 2.2% (b) 2.4%
(c) 4.2% (d) 4.4%
(e) None of these

[IBPS PO/MT Examination, 2012]

96. What is the total cost of renovation of the hall for customer transaction and the locker area?

(a) ₹2,29,100 (b) ₹2,30,206
(c) ₹2,16,920 (d) ₹2,42,440
(e) None of these

[IBPS PO/MT, 2012]

97. Water is flowing at the rate of 5 Km/h through a pipe of diameter 14 cm into a rectangular tank which is 50 m long, 44 m wide. The time taken, in hours, for the rise in the level of water in the tank to be 7 cm is:

(a) 2 (b) $1\frac{1}{2}$
(c) 3 (d) $2\frac{1}{2}$

[SSC (GL) Examination, 2011]

98. The areas of three consecutive faces of a cuboid are 12 cm^2, 20 cm^2 and 15 cm^2, then the volume (in cm^3) of the cuboid is:

(a) 3600 (b) 100
(c) 80 (d) 60

[SSC (GL) Examination, 2011]

99. Water is flowing at the rate of 3 Km/h through a circular pipe of 20 cm internal diameter into a circular cistern of diameter 10 m and depth 2 m. In how much time will the cistern be filled?

(a) 1 hour (b) 1 hour 40 mins
(c) 1 hours 20 mins (d) 2 hours 40 mins

[SSC (GL) Examination, 2011]

100. Marbles of diameter 1.4 cm are dropped into a cylindrical beaker containing some water and are fully submerged. The diameter of the beaker is 7 cm. Find how many marbles have been dropped in it if the water rises by 5.6 cm.

(a) 50 (b) 150
(c) 250 (d) 350

[SSC (GL) Examination, 2011]

101. A hemisphere and a cone have equal bases. If their heights are also equal, the ratio of their curved surfaces will be:

(a) $1:\sqrt{2}$ (b) $2:\sqrt{2}$
(c) 1:2 (d) 2:1

[SSC (GL) Examination, 2011]

102. The heights of a cone, cylinder and hemisphere are equal. If their radii are in the ratio 2:3:1, then the ratio of their volumes is:

(a) 2:9:2 (b) 4:9:1
(c) 4:27:2 (d) 2:3:1

[SSC Examination, 2011]

103. Base of a right pyramid is a square, length of diagonal of the base is $24\sqrt{2}$ m. If the volume of the pyramid is 1728 cu. m, its height is:

(a) 7 m (b) 8 m
(c) 9 m (d) 10 m

[SSC Examination, 2011]

104. The height of a right circular cone and the radius of its circular base are 9 cm and 3 cm respectively. The cone is cut by a plane parallel to its base so as to divide it into two parts. The volume of the frustum (i.e., the lower part) of the cone is 44 cm^2. The radius of the upper circular surface of the frustum $\left(\text{taking } \pi = \frac{22}{7}\right)$ is:

(a) $\sqrt[3]{12}$ cm (b) $\sqrt[3]{13}$ cm
(c) $\sqrt[3]{6}$ cm (d) $\sqrt[3]{20}$ cm

[SSC Examination, 2011]

105. The ratio of radii of 2 right circular cylinders is 2:3 and their heights are in the ratio 5:4. The ratio of their curved surface area is:

(a) 5:6 (b) 3:4
(c) 4:5 (d) 2:3

[SSC Examination, 2011]

106. A solid cylinder has total surface area of 462 sq.cm. Curved surface area is $\frac{1}{3}$rd of its total surface area. The volume of the cylinder is:

(a) 530 cm^2 (b) 536 cm^2
(c) 539 cm^2 (d) 545 cm^2

[SSC Examination, 2011]

107. A cylinder and a cone have equal radii of their bases and equal heights. If their curved surface areas are in the ratio 8:5, the ratio of their radius and height is:

(a) 1:2 (b) 1:3
(c) 2:3 (d) 3:4

[SSC Examination, 2011]

108. A solid is hemispherical at the bottom and conical above. If the surface areas of the two parts are equal, then the ratio of radius and height of its conical part is:

(a) 1:3 (b) 1:1
(c) 3:1 (d) $1:\sqrt{3}$

[SSC Examination, 2011]

109. Base of a right prism is an equilateral triangle of side 6 cm. If the volume of the prism is $108\sqrt{3}$ cc, its height is:

(a) 9 cm (b) 10 cm
(c) 11 cm (d) 12 cm

[SSC Examination, 2011]

110. The number of coins, each of radius 0.75 cm and thickness 0.2 cm, to be melted to make a right circular cylinder of height 8 cm and radius 3 cm, is:

(a) 640 (b) 600
(c) 500 (d) 480

[SSC Examination, 2010]

111. If the radius of a sphere is increased by 2 m, its surface area is increased by 704 m^2. What isthe radius of the original sphere? $\left(\text{use } \pi = \frac{22}{7}\right)$

(a) 16 m^2 (b) 15 m^2
(c) 14 m^2 (d) 13 m^2

[SSC Examination, 2010]

112. A right circular cylinderis circumscribing a hemisphere such that their bases are common. The ratio of their volumes is.

(a) 1:3 (b) 1:2
(c) 2:3 (d) 3:4

[SSC Examination, 2010]

113. 3 spherical balls of radii 1 cm, 2 cm and 3 cm are melted to form a single spherical ball. In the process, the loss of material is 25%. The radius of the new ball is:

(a) 6 cm (b) 5 cm
(c) 3 cm (d) 2 cm

[SSC Examination, 2010]

114. The length of the diagonal of a cube is 6 cm. The volume of the cube (in cm^3) is:

(a) $18\sqrt{3}$ (b) $24\sqrt{3}$
(c) $28\sqrt{3}$ (d) $30\sqrt{3}$

[SSC Examination, 2010]

115. If a sphere of radius r is divided into 4 identical parts, then the total surface area of the 4 parts is:

(a) $4\pi r^2$ square units (b) $2\pi r^2$ square units
(c) $8\pi r^2$ square units (d) $3\pi r^2$ square units

[SSC Examination, 2010]

ANSWER KEYS

EXERCISE-1

1. (c)	**2.** (b)	**3.** (c)	**4.** (b)	**5.** (c)	**6.** (b)	**7.** (c)	**8.** (c)	**9.** (b)	**10.** (a)	**11.** (d)	**12.** (b)
13. (b)	**14.** (c)	**15.** (c)	**16.** (c)	**17.** (b)	**18.** (a)	**19.** (d)	**20.** (c)	**21.** (b)	**22.** (d)	**23.** (c)	**24.** (c)
25. (c)	**26.** (a)	**27.** (d)	**28.** (a)	**29.** (c)	**30.** (c)	**31.** (d)	**32.** (b)	**33.** (d)	**34.** (a)	**35.** (b)	**36.** (a)
37. (a)	**38.** (d)	**39.** (d)	**40.** (b)	**41.** (a)	**42.** (c)	**43.** (b)	**44.** (c)	**45.** (a)	**46.** (a)	**47.** (c)	**48.** (d)
49. (a)	**50.** (a)	**51.** (a)	**52.** (b)	**53.** (d)	**54.** (b)	**55.** (d)	**56.** (b)	**57.** (d)	**58.** (c)	**59.** (d)	**60.** (b)
61. (a)	**62.** (b)	**63.** (b)	**64.** (b)	**65.** (b)	**66.** (a)	**67.** (a)	**68.** (c)	**69.** (b)	**70.** (d)	**71.** (b)	**72.** (c)
73. (c)	**74.** (a)	**75.** (d)	**76.** (b)	**77.** (b)	**78.** (c)	**79.** (a)	**80.** (b)	**81.** (d)	**82.** (b)	**83.** (a)	**84.** (c)
85. (c)	**86.** (b)	**87.** (a)	**88.** (c)	**89.** (b)	**90.** (c)	**91.** (b)	**92.** (a)	**93.** (b)	**94.** (a)	**95.** (c)	**96.** (a)
97. (b)	**98.** (c)	**99.** (c)	**100.** (c)	**101.** (a)	**102.** (d)	**103.** (a)	**104.** (b)	**105.** (a)	**106.** (c)	**107.** (a)	**108.** (b)
109. (d)	**110.** (c)	**111.** (c)	**112.** (b)	**113.** (c)	**114.** (c)	**115.** (b)					

EXERCISE-2

1. (b)	**2.** (b)	**3.** (d)	**4.** (d)	**5.** (b)	**6.** (d)	**7.** (b)	**8.** (a)	**9.** (a)	**10.** (b)	**11.** (b)	**12.** (c)
13. (d)	**14.** (b)	**15.** (c)	**16.** (c)	**17.** (b)	**18.** (a)	**19.** (a)	**20.** (b)	**21.** (b)	**22.** (b)	**23.** (b)	**24.** (d)
25. (a)	**26.** (b)	**27.** (b)	**28.** (d)	**29.** (a)	**30.** (c)	**31.** (a)	**32.** (b)	**33.** (a)	**34.** (d)	**35.** (c)	**36.** (c)
37. (b)	**38.** (d)	**39.** (a)	**40.** (b)	**41.** (b)	**42.** (c)	**43.** (a)	**44.** (d)	**45.** (d)	**46.** (a)	**47.** (b)	**48.** (c)
49. (a)	**50.** (c)	**51.** (a)	**52.** (d)	**53.** (d)	**54.** (a)	**55.** (c)	**56.** (e)	**57.** (a)	**58.** (a)	**59.** (d)	**60.** (c)
61. (a)	**62.** (d)	**63.** (c)	**64.** (d)	**65.** (d)	**66.** (d)	**67.** (a)	**68.** (a)	**69.** (a)	**70.** (a)	**71.** (c)	**72.** (d)
73. (d)	**74.** (b)	**75.** (a)	**76.** (c)	**77.** (d)	**78.** (d)	**79.** (c)	**80.** (c)	**81.** (a)	**82.** (c)	**83.** (c)	**84.** (a)
85. (c)	**86.** (b)	**87.** (c)	**88.** (a)	**89.** (b)	**90.** (c)	**91.** (a)	**92.** (c)	**93.** (e)	**94.** (e)	**95.** (b)	**96.** (a)
97. (a)	**98.** (d)	**99.** (b)	**100.** (b)	**101.** (b)	**102.** (c)	**103.** (c)	**104.** (b)	**105.** (a)	**106.** (c)	**107.** (d)	**108.** (d)
109. (d)	**110.** (a)	**111.** (d)	**112.** (c)	**113.** (c)	**114.** (b)	**115.** (c)					